Climate Change Governance: Science, Data & Models

Lecture 2: Radiations & Greenhouse gases

ENGG562/ENV562 – Fall 2026

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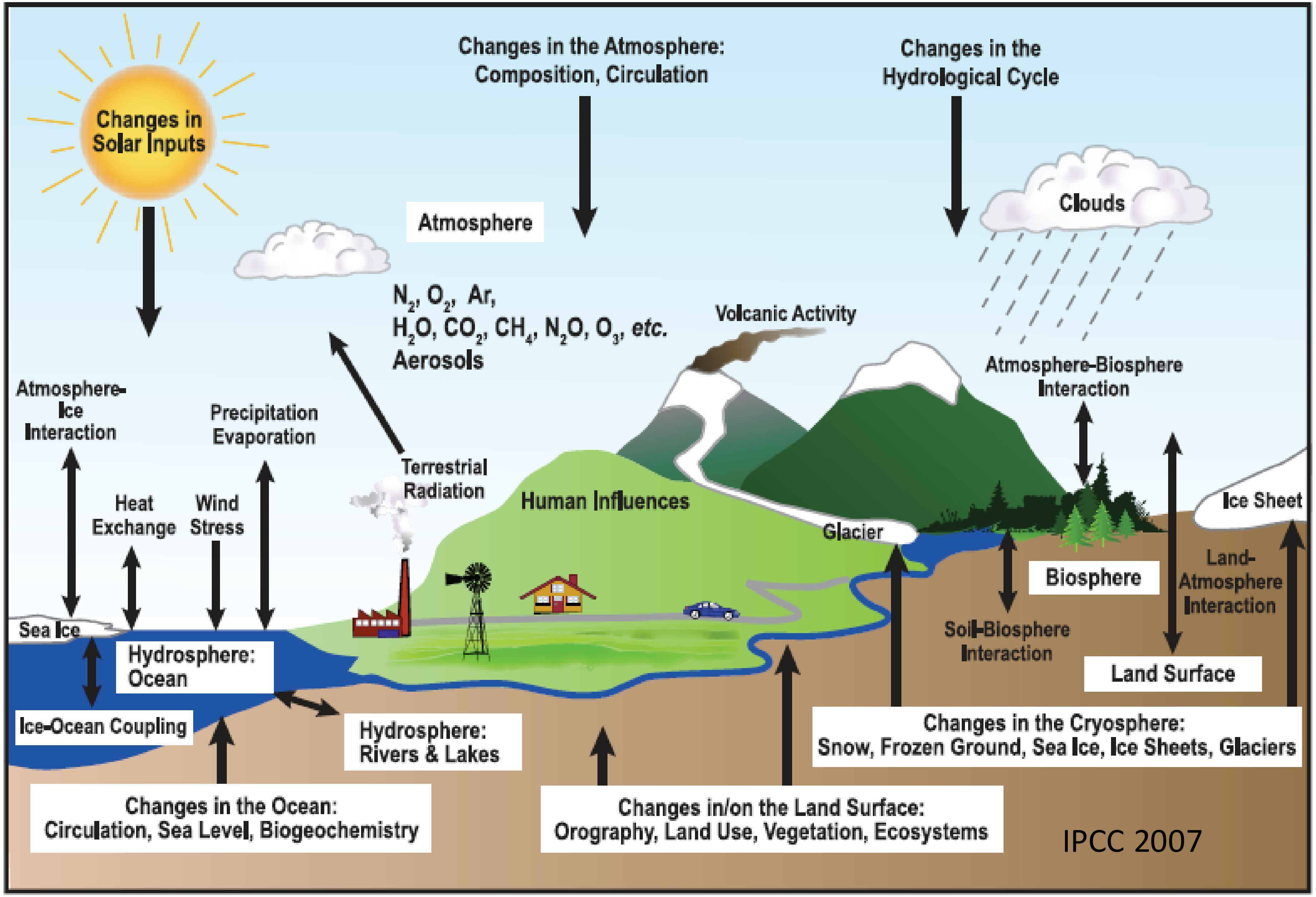
Recap: Earth System

- **Boxes** store atmosphere, ocean, land, ice, biosphere
- **Arrows** exchange

Every arrow carries one of four things:

- **energy**
- **water**
- **momentum**
- **carbon**



What’s a budget equation?

rate-of-change = input minus output

$$\Delta X = X_2 - X_1$$
$$\Delta t = t_2 - t_1$$

$$\xrightarrow[\text{Input}]{I} \boxed{\frac{\Delta X}{\Delta t} = I - O} \xrightarrow[\text{Output}]{O}$$

Units: something/time

Example:

In: `RA stipend 60,000 + parents 20,000 → I = 80,000 PKR/month`

Out: `hostel 35,000 + food 20,000 + transport & phone 8,000 + books 5,000 → O = 68,000 PKR/month`

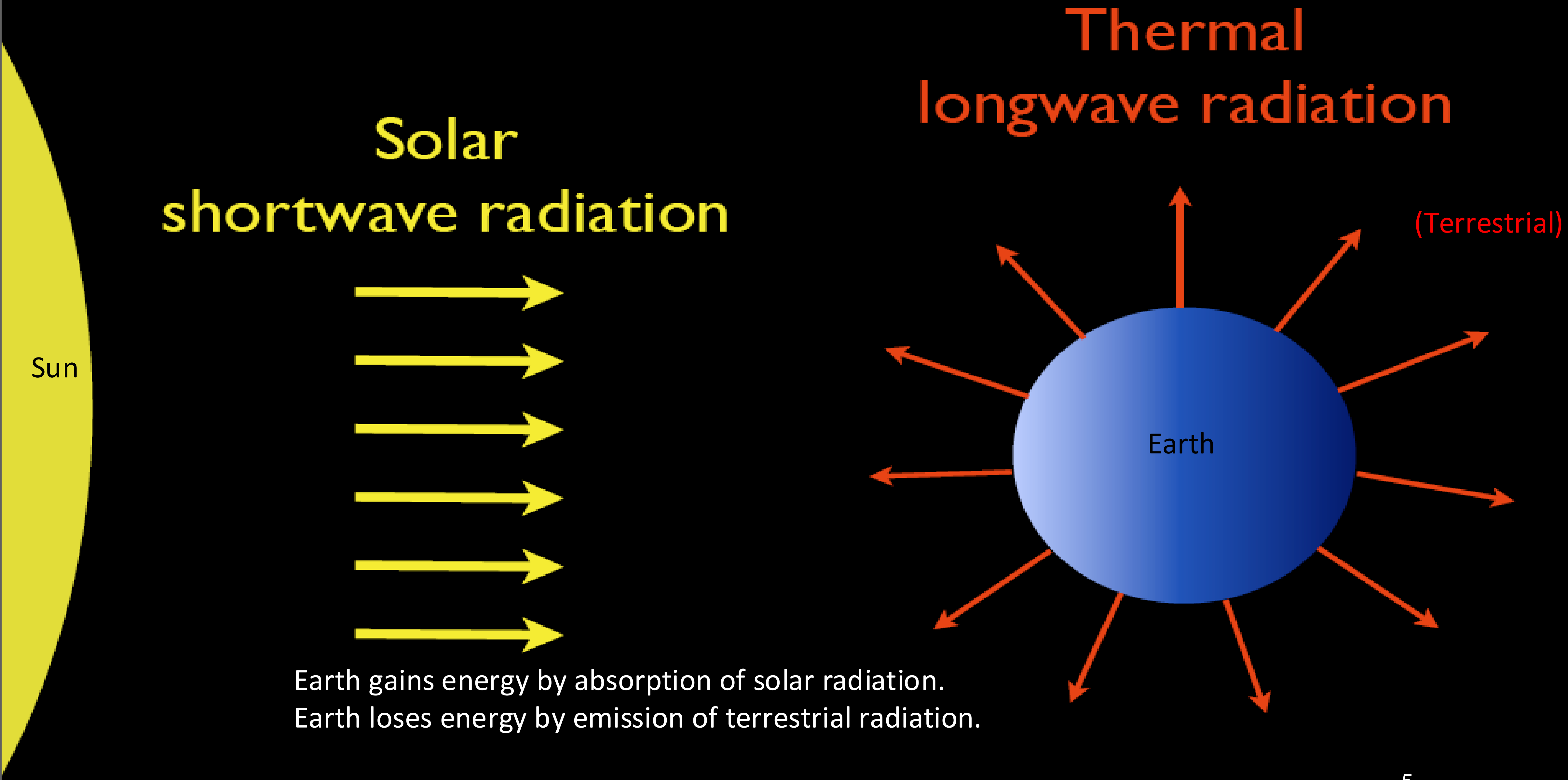
ΔX `= 80,000 – 68,000 = +12,000 PKR/month (savings grow)`

What's a budget in balance?

- A budget in balance is one that doesn't change
- Input = Output

$$\Delta X = 0$$

- What's Earth's energy source?
- What's Earth's energy sink?

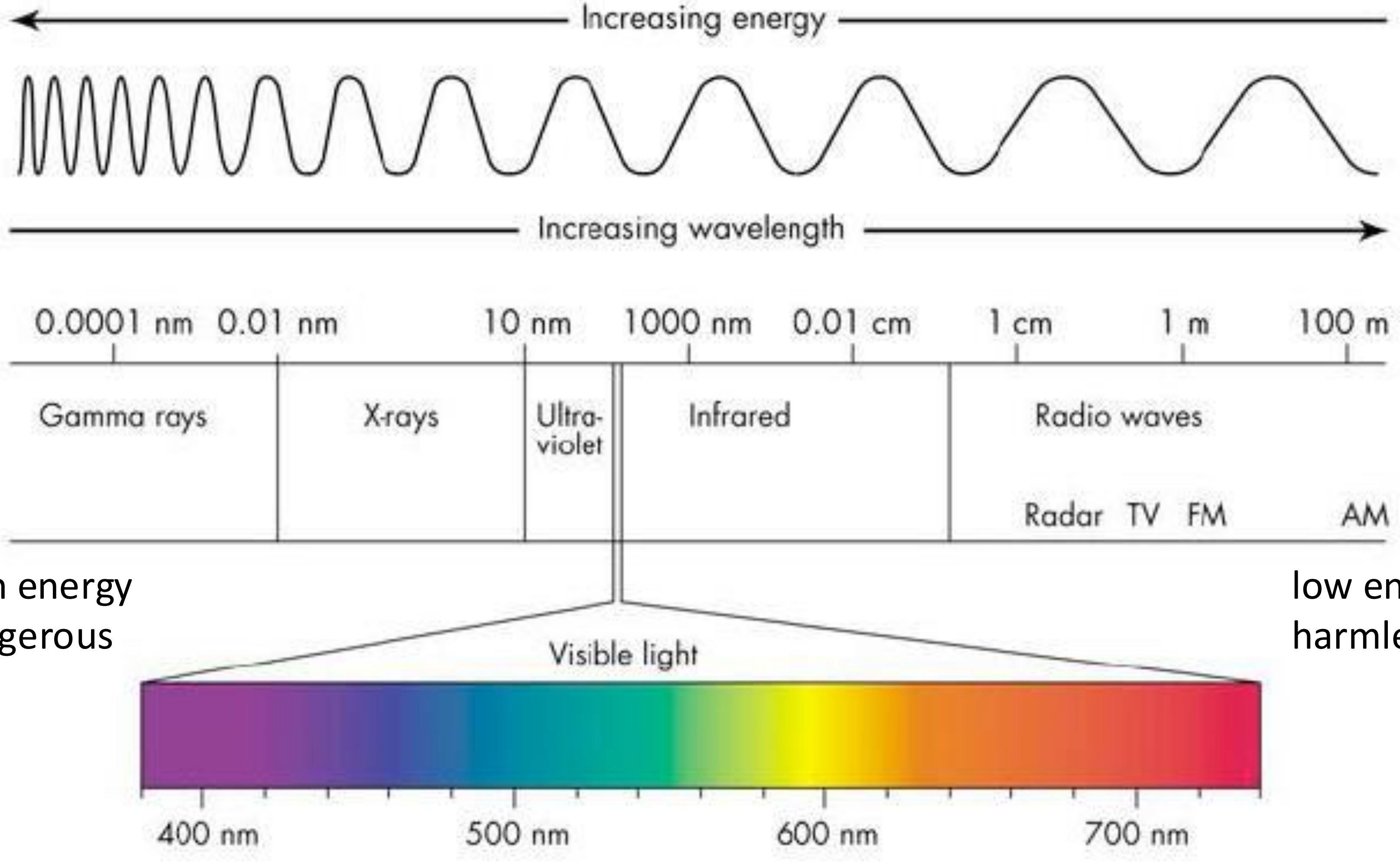


Earth gains energy by absorption of solar radiation.
Earth loses energy by emission of terrestrial radiation.

Why is radiation from the sun emitted at shorter wavelengths than that emitted from Earth?

Einstein-Planck relation: E= $E = h\nu = hc/\lambda$ *explains the single photon*

Electromagnetic Spectrum

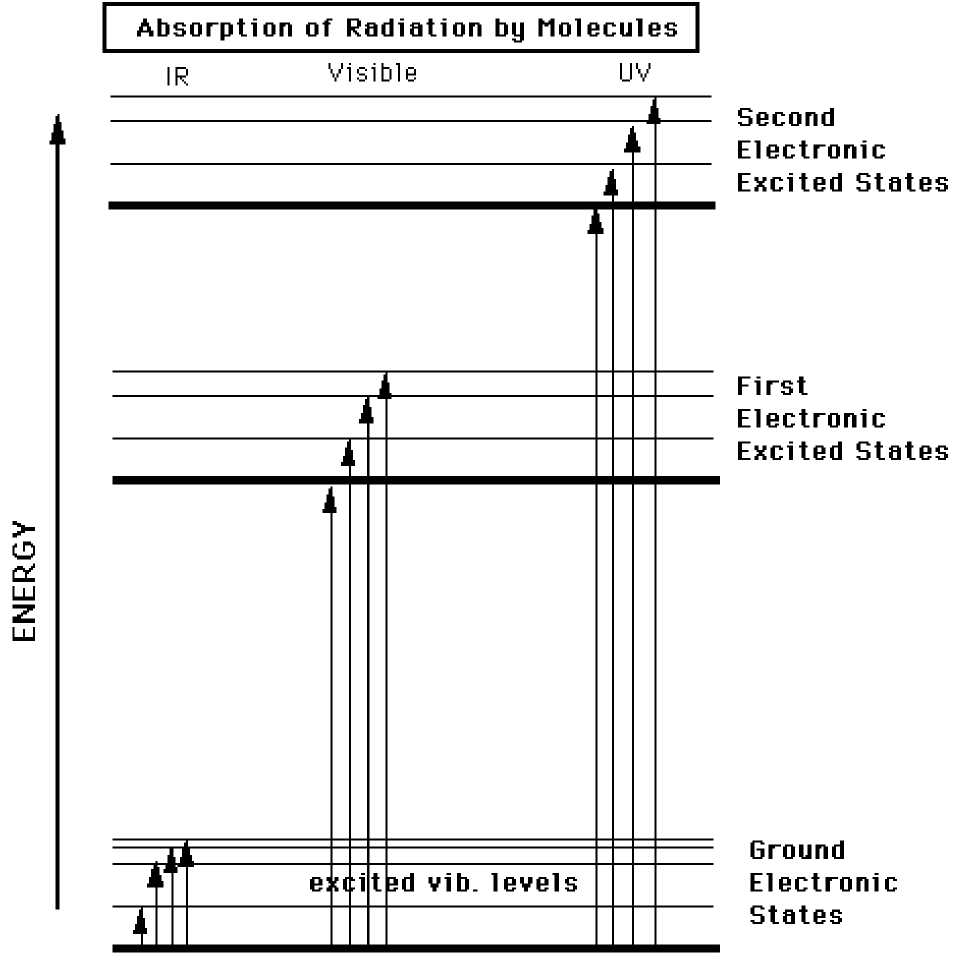


high energy
dangerous

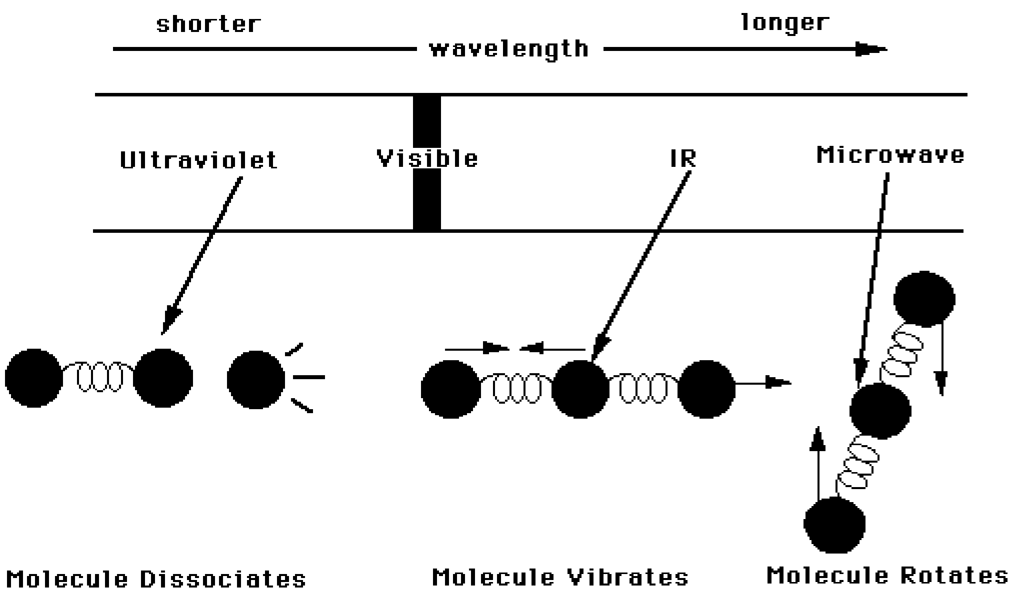
low energy
harmless

1 nm = 10^{-9}m = 0.0000000001 m = 10^{-6} mm = 10^{-3} µm

Interaction of electromagnetic radiation with matter



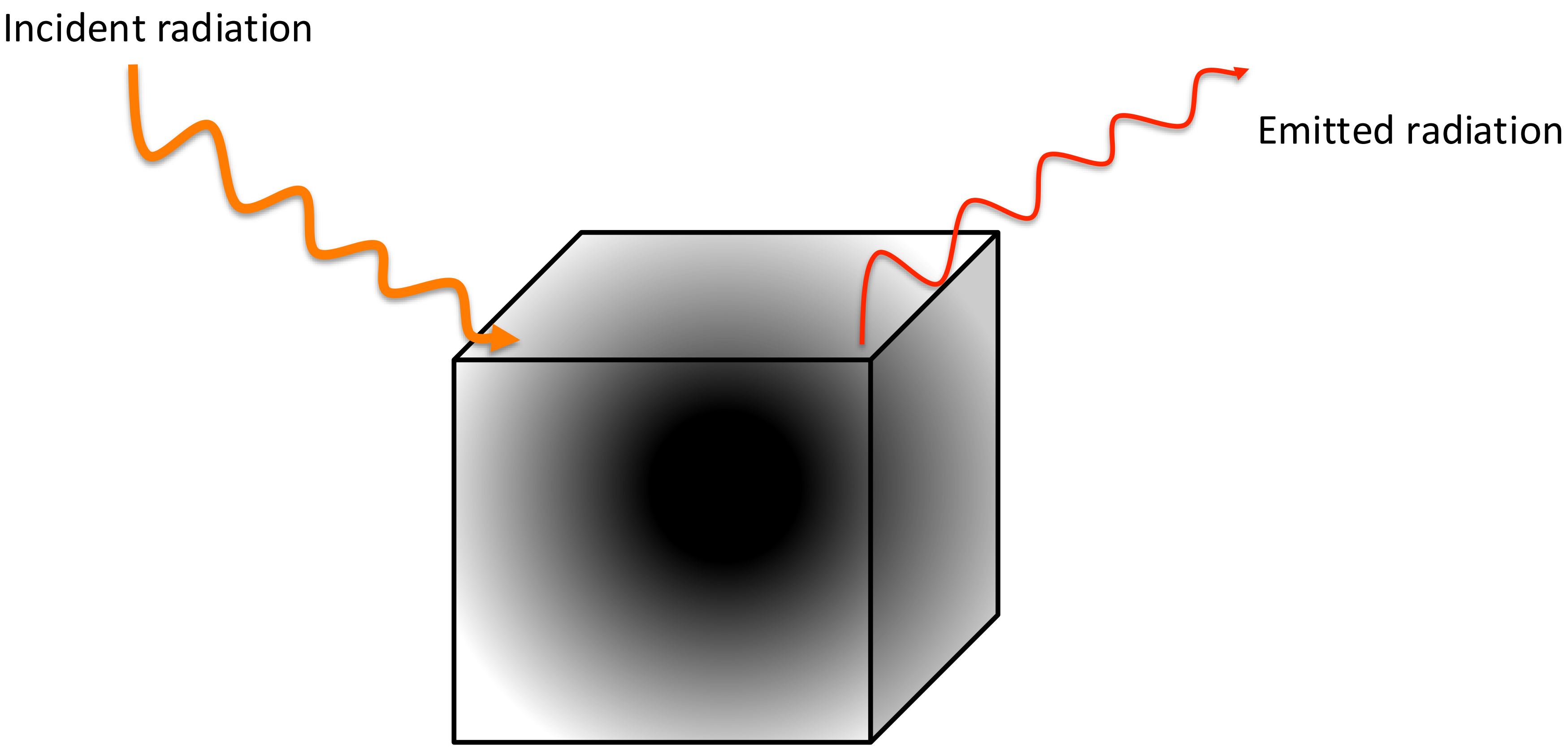
Each excited electronic energy state has sub-states with different vibrational levels.



Absorption : transition from a higher to lower energy state

Emission : release of a photon from a lower to higher energy state

Black body radiation



• A ‘**black body**’ is an ‘idealized’ object completely absorbing all incident electromagnetic radiation no matter what its wavelength (frequency). It emits thermal radiation according to Planck’s law.

• Many natural objects emit thermal radiation like a black body (*even fresh snow or the sun*). But we’ll discuss some exceptions later.

Black body radiation

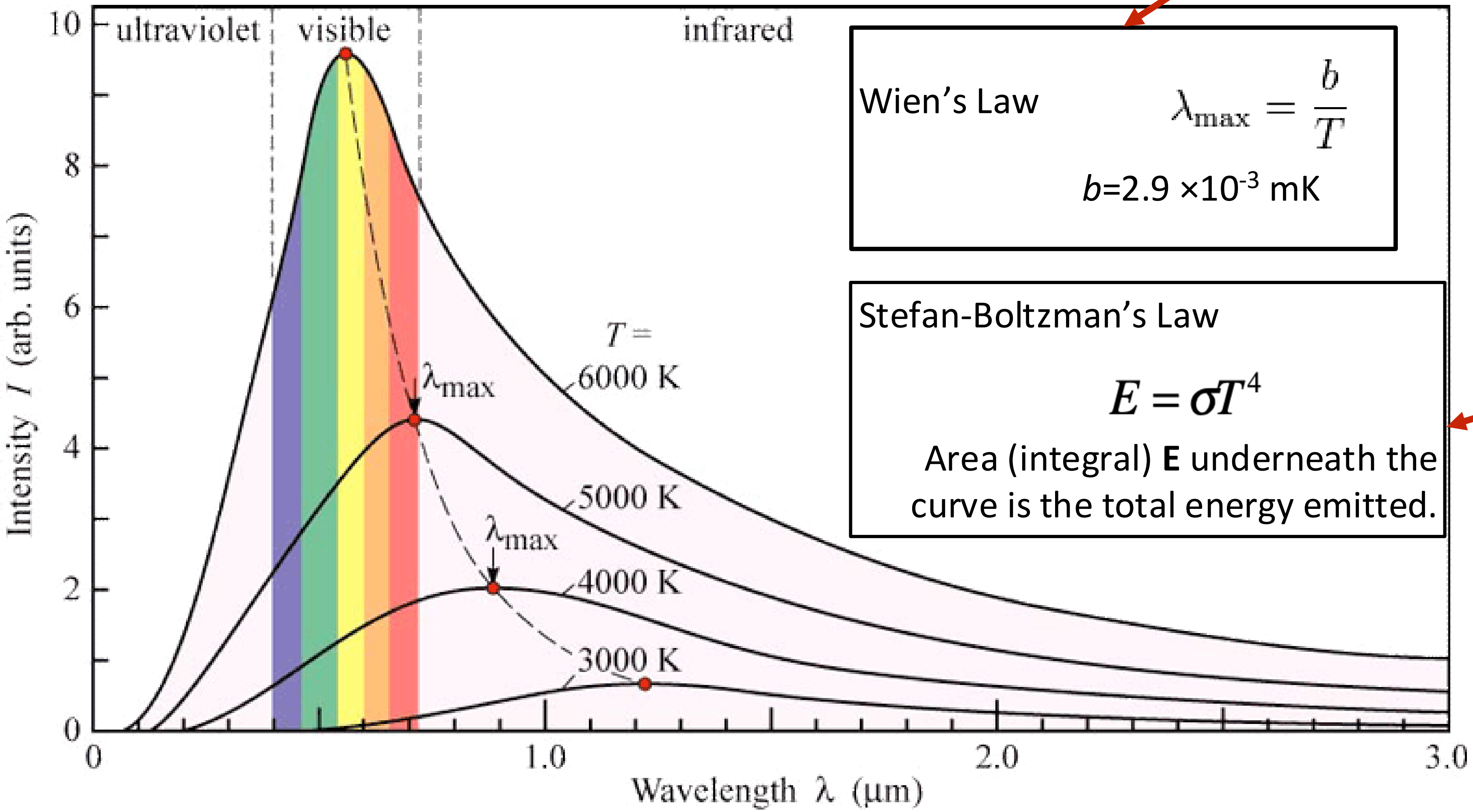
- We cannot see the radiation emitted from an object at room temperature, because it is in the infrared part of the spectrum, but we can feel it.
- At hot temperatures objects emit visible radiation and start to glow (incandescence).

Black body radiation

Shape of Curves → Planck's Law

$$I'(\lambda, T) = \frac{2hc^2}{\lambda^5}\frac{1}{e^{\frac{hc}{\lambda kT}} - 1}.$$

Location of peak



Wien's Law $\lambda_{\max} = \frac{b}{T}$

b=2.9 ×10^{-3} mK

Stefan-Boltzman's Law

$$E = \sigma T^4$$

Area (integral) **E** underneath the curve is the total energy emitted.

Area under the curve

Sun's temperature 5000 - 6000 K

How much total radiation does an object emit ?

Stefan-Boltzmann Law

$$E = \varepsilon\sigma T^4$$

where: σ is a constant equal to 5.67×10^{-8} [W/ (m^2 K^4)] (Stefan Boltzman constant)

E is energy flux in units of [W / m^2]

T is absolute temperature in units of Kelvin [K]

ε is material-specific constant (emissivity), range 0.6 to 1

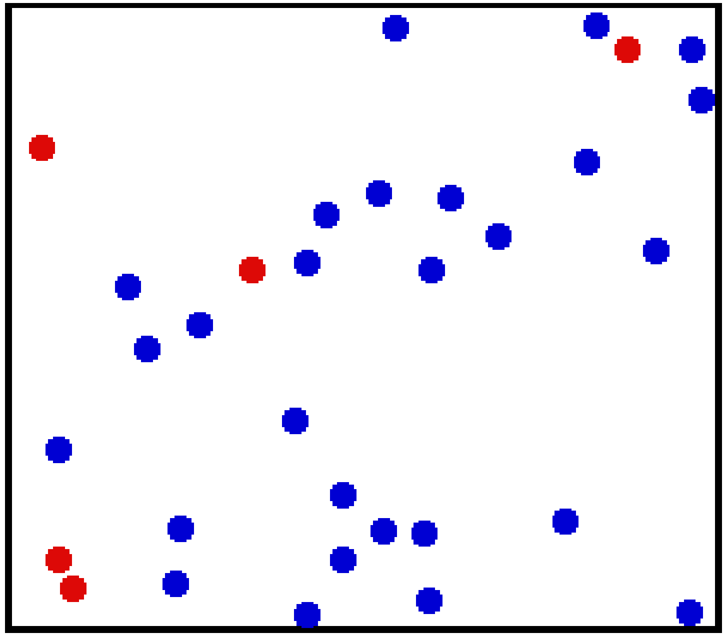
(for most materials it is close to 1 (e.g. snow has $\varepsilon > 0.94$)

Note: If you double the temperature of an object its radiative energy output increases by $2^4 = 16$ times ! *Careful, you need to use absolute (Kelvin) temperature to get the numbers right.*

Stefan-Boltzman Law

Temperature

(Third law of thermodynamics)



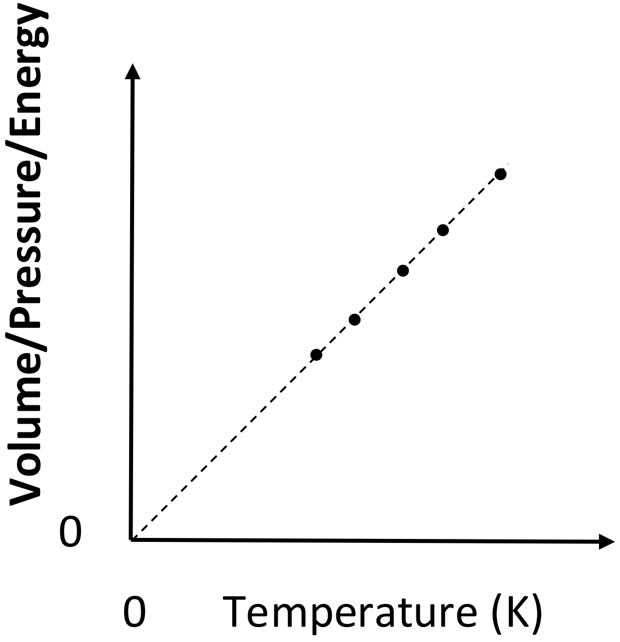
Definition of absolute temperature scale.

T = 0 K at –273°C or –460°F.

T = 273 K at 0°C or 32°F, the freezing point of water.

T = 373 K at 100°C or 212°F, the boiling point of water.

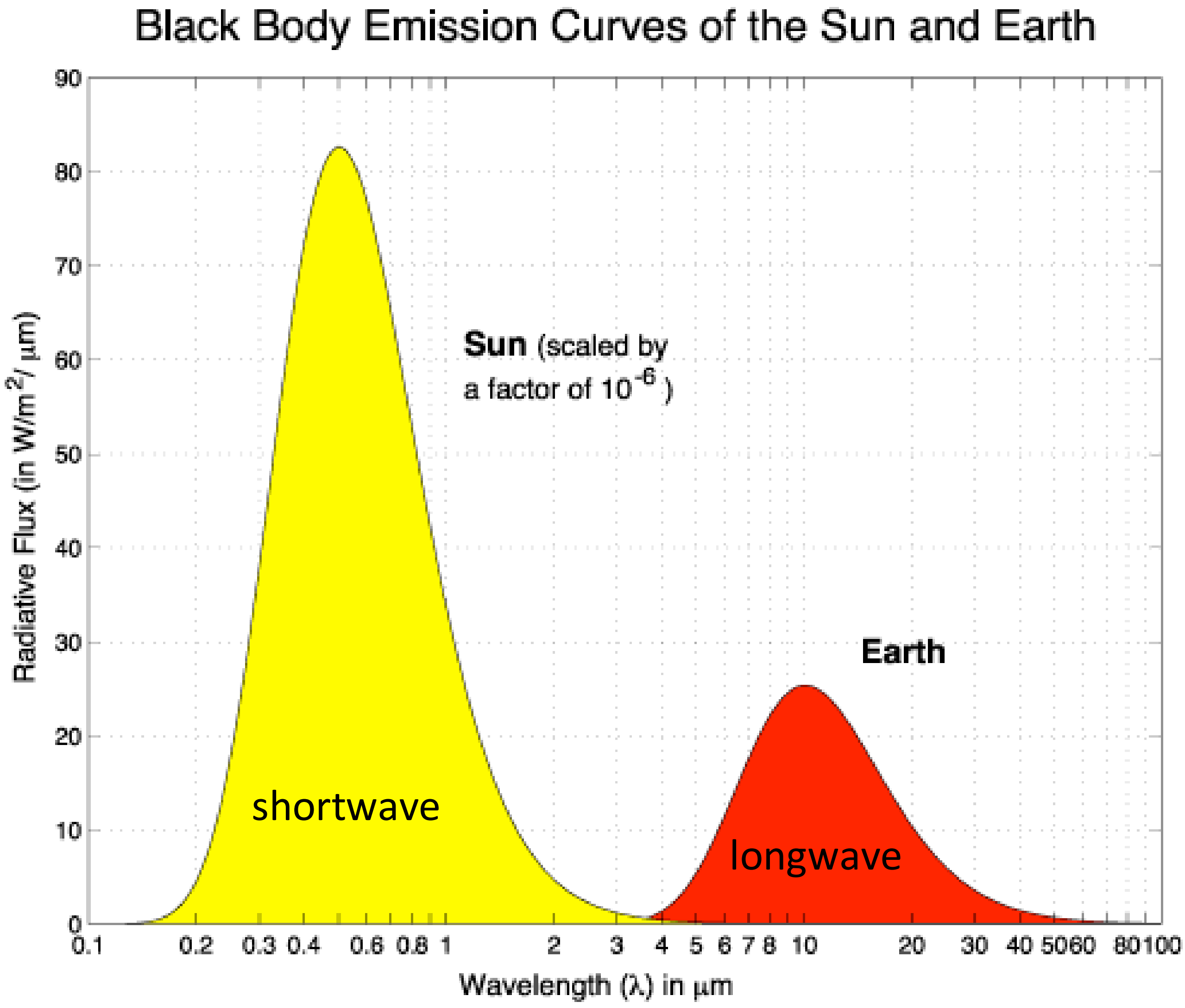
Temperature differences: 1 K = 1°C = 1.8°F



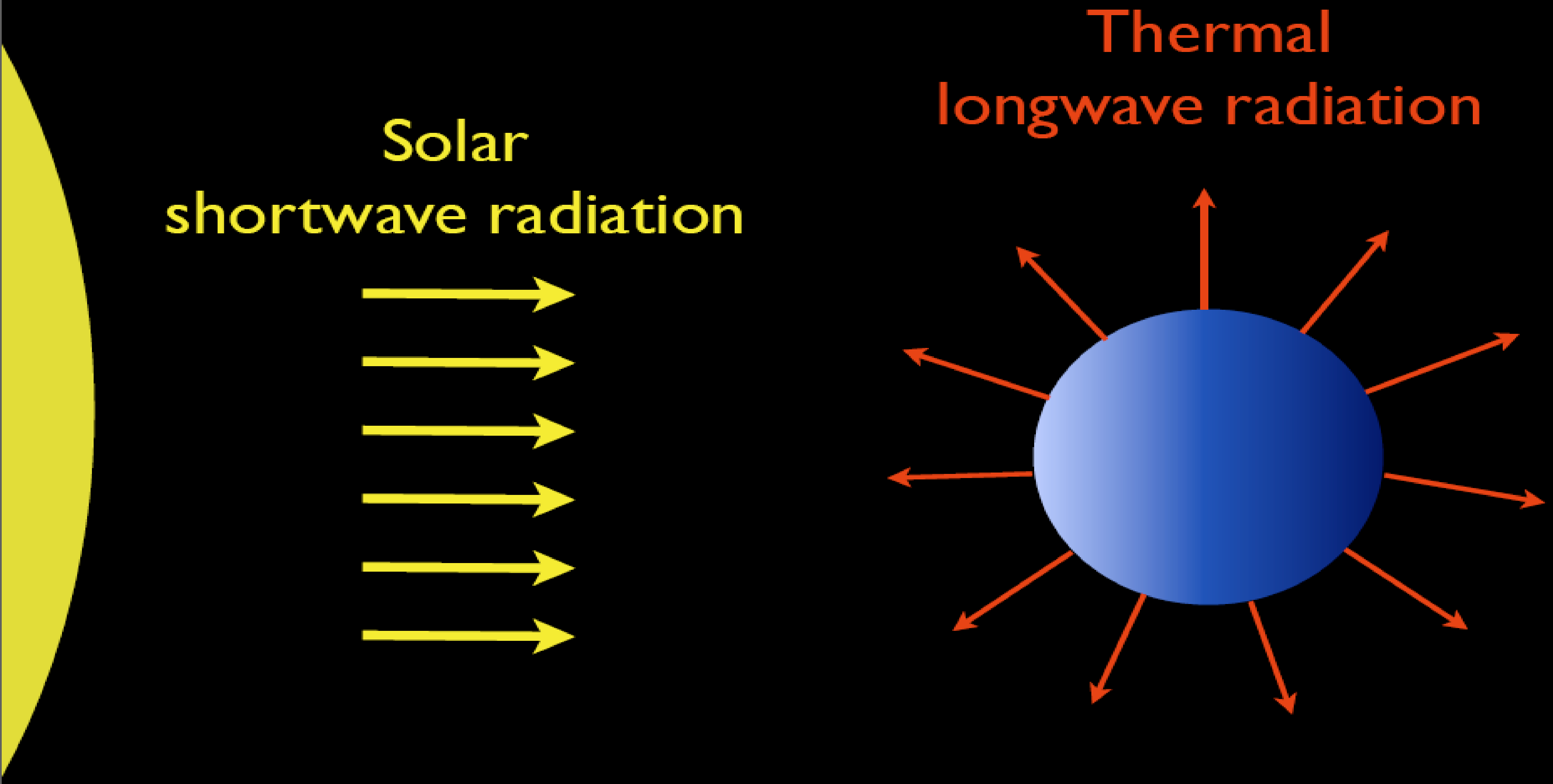
- Temperature is caused by the kinetic energy (random motions) of molecules.
- At absolute zero temperature T = 0 K, all molecular motion ceases.
- It is theoretically impossible to be achieved, but scientists have gotten pretty close.
- An ideal gas would have zero volume, zero pressure, and zero energy at T = 0 K.

Planck's, Stefan-Boltzman's & Wien's Laws, summary

Emission for Sun and Earth Temperatures

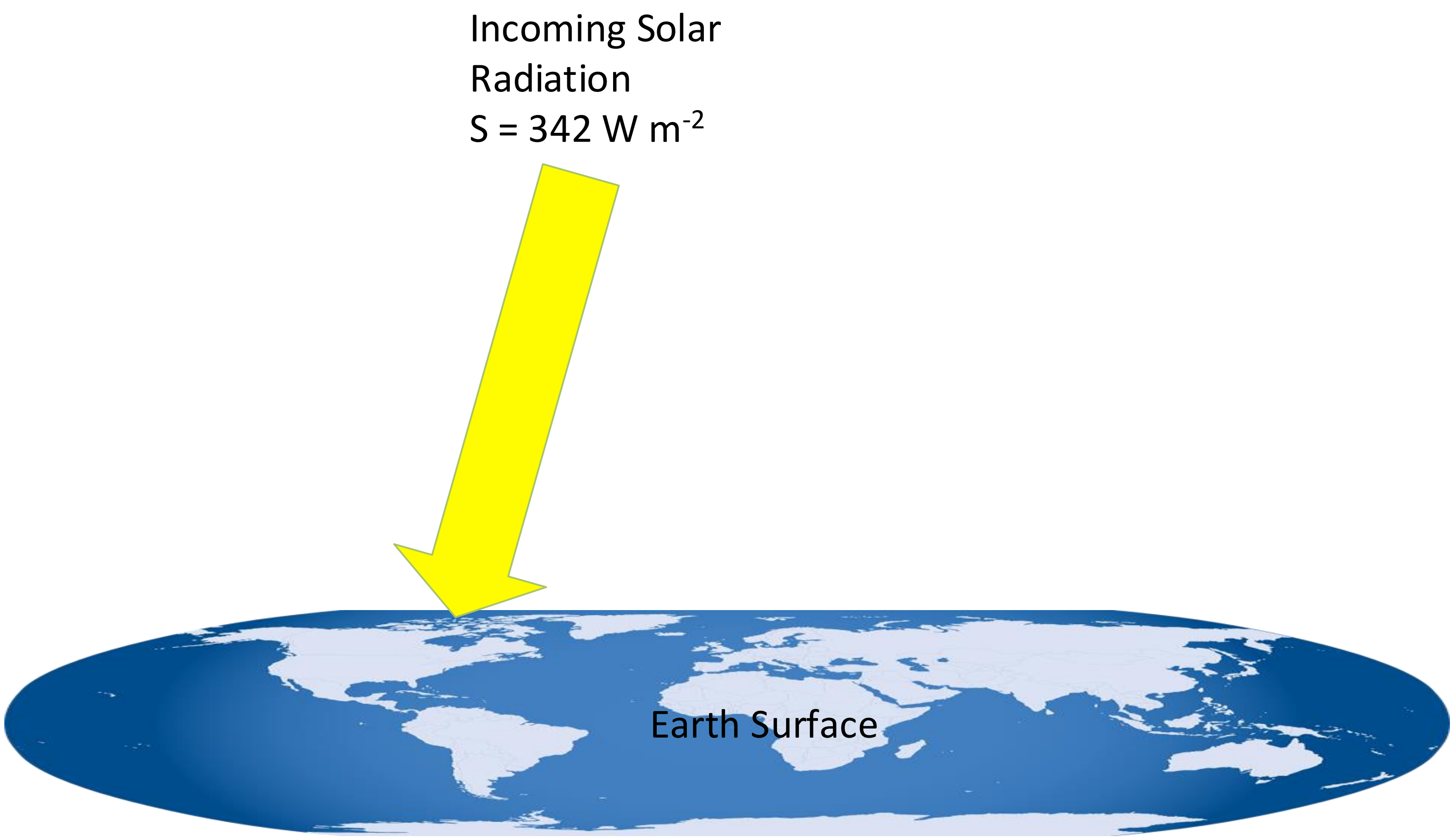


- Light has color. Color is given by wavelength.
- The higher the temperature, the greater the rate of emission.
- The higher the temperature, the shorter the wavelength of the peak in emission.
- Little overlap between solar and terrestrial radiation



Total Solar Irradiance TSI = 1,370 W/m^2(perpendicular to sun’s rays)
Averaged over surface area of Earth S_0 = TSI/4 = 342 W/m^2 since surface area of sphere ($A=4\pi R^2$) is 4 times the surface area of a disc ($A=\pi R^2$) with the same radius R.

Energy Balance Model 1 (Bare Rock)



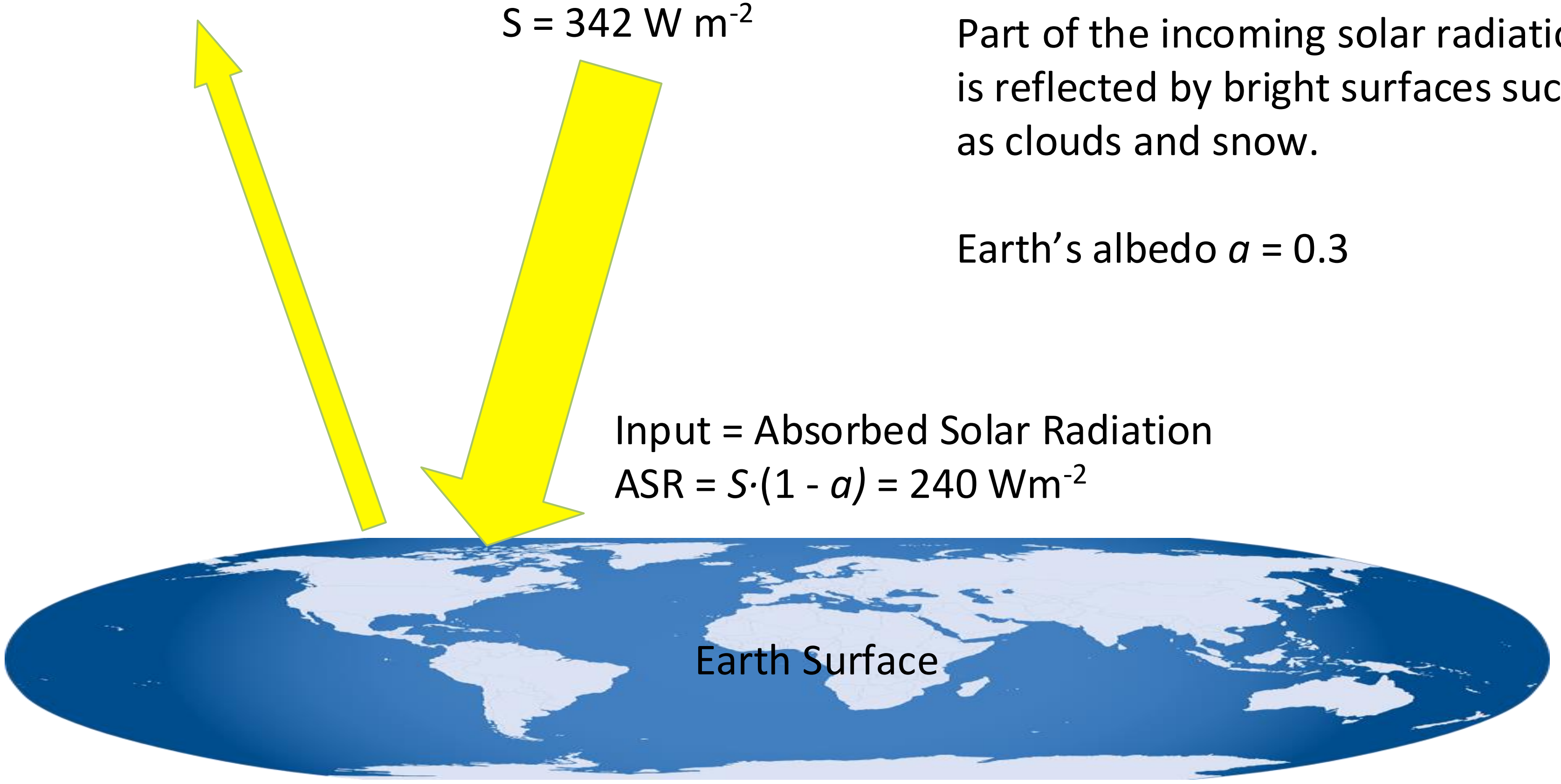
Energy Balance Model 1 (Bare Rock)

Reflected Solar
$S \cdot a$ = 102 W m^{-2}

Incoming Solar
Radiation
S = 342 W m^{-2}

Part of the incoming solar radiation is reflected by bright surfaces such as clouds and snow.

Earth's albedo a = 0.3



Input = Absorbed Solar Radiation
ASR = $S \cdot (1 - a)$ = 240 Wm^{-2}

Energy Balance Model 1 (Bare Rock)

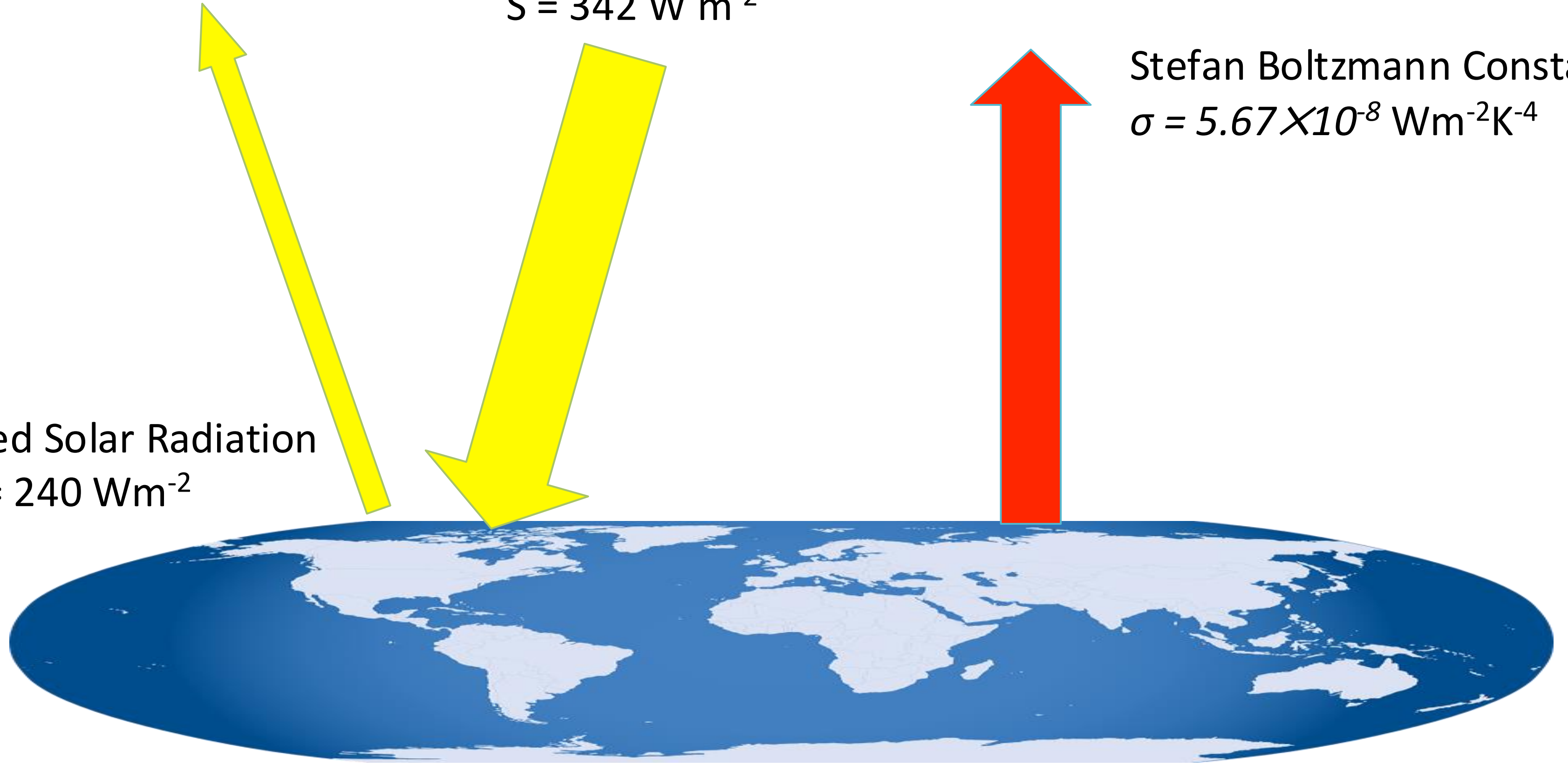
Reflected Solar
$S \cdot a = 102\ \mathrm{W\ m^{-2}}$

Incoming Solar Radiation
$S = 342\ \mathrm{W\ m^{-2}}$

Output = Emitted Terrestrial Radiation
$ETR = \sigma T^4$

Stefan Boltzmann Constant
$\sigma = 5.67 \times 10^{-8}\ \mathrm{Wm^{-2}K^{-4}}$

Input = Absorbed Solar Radiation
$\mathrm{ASR} = S \cdot (1 - a) = 240\ \mathrm{Wm^{-2}}$



Input = Output
$ASR = ETR$
$S \cdot (1 - a) = \sigma T^4$ → $\mathrm{T} = ?$

Calculate Earth's surface temperature!

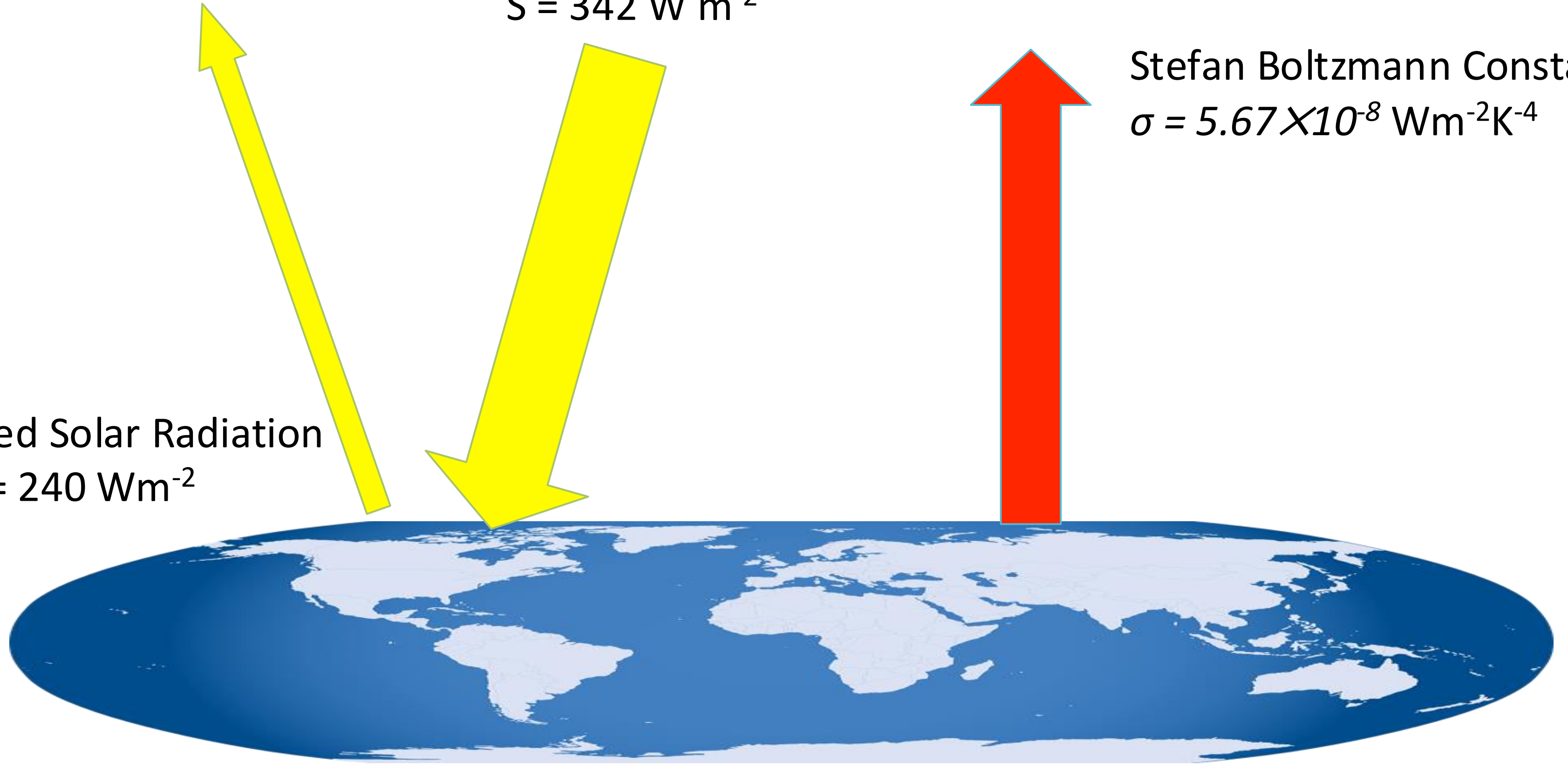
Energy Balance Model 1 (Bare Rock)

Reflected Solar
$S \cdot a$ = 102 W m^{-2}

Incoming Solar Radiation
S = 342 W m^{-2}

Output = Emitted Terrestrial Radiation
$ETR = \sigma T^4$

Stefan Boltzmann Constant
$\sigma = 5.67 \times 10^{-8}$ Wm^{-2}K^{-4}



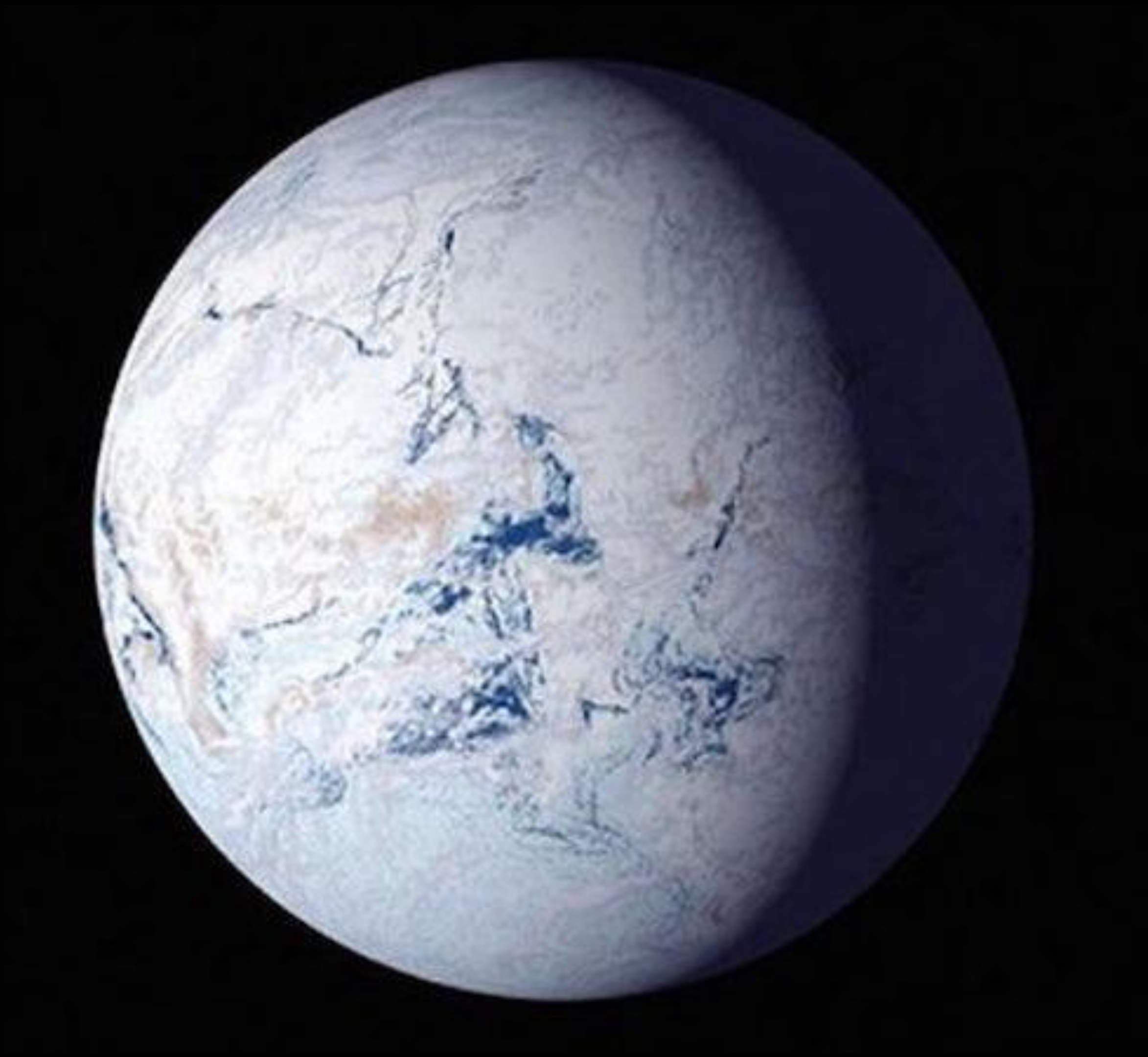
Input = Absorbed Solar Radiation
ASR = $S \cdot (1 - a)$ = 240 Wm^{-2}

Input = Output
$ASR = ETR$
$S \cdot (1 - a) = \sigma T^4$ $\longrightarrow$ $T = (ASR/\sigma)^{1/4}$ = 255 K = -18°C = 0°F

Something is wrong …

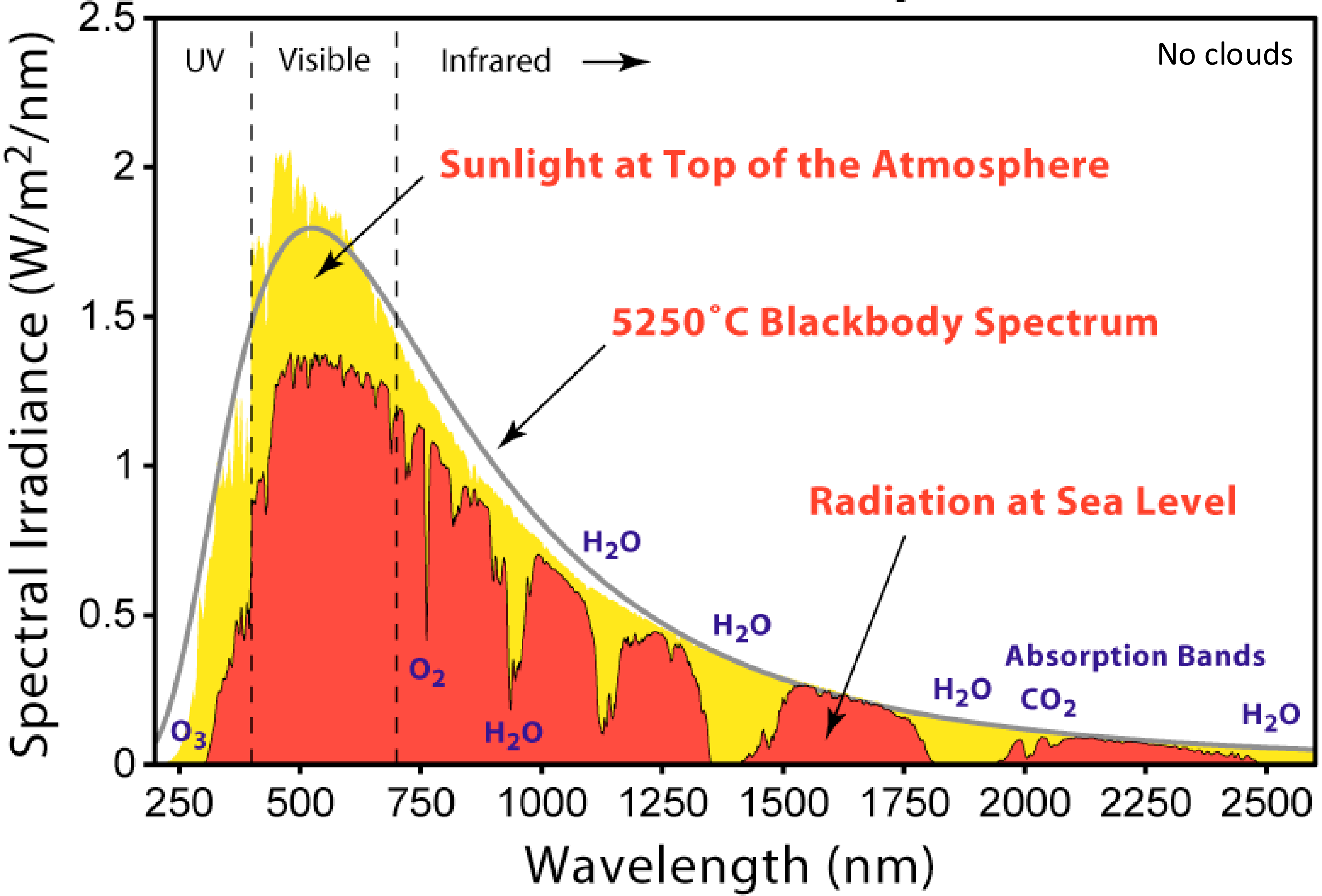
The atmosphere is missing! It acts like a blanket or the glass of a greenhouse and keeps surface temperatures warmer than they would be otherwise. Our bare rock model is good for Mars, which has a thin atmosphere, but not for planets with thick atmospheres like Earth or Venus.

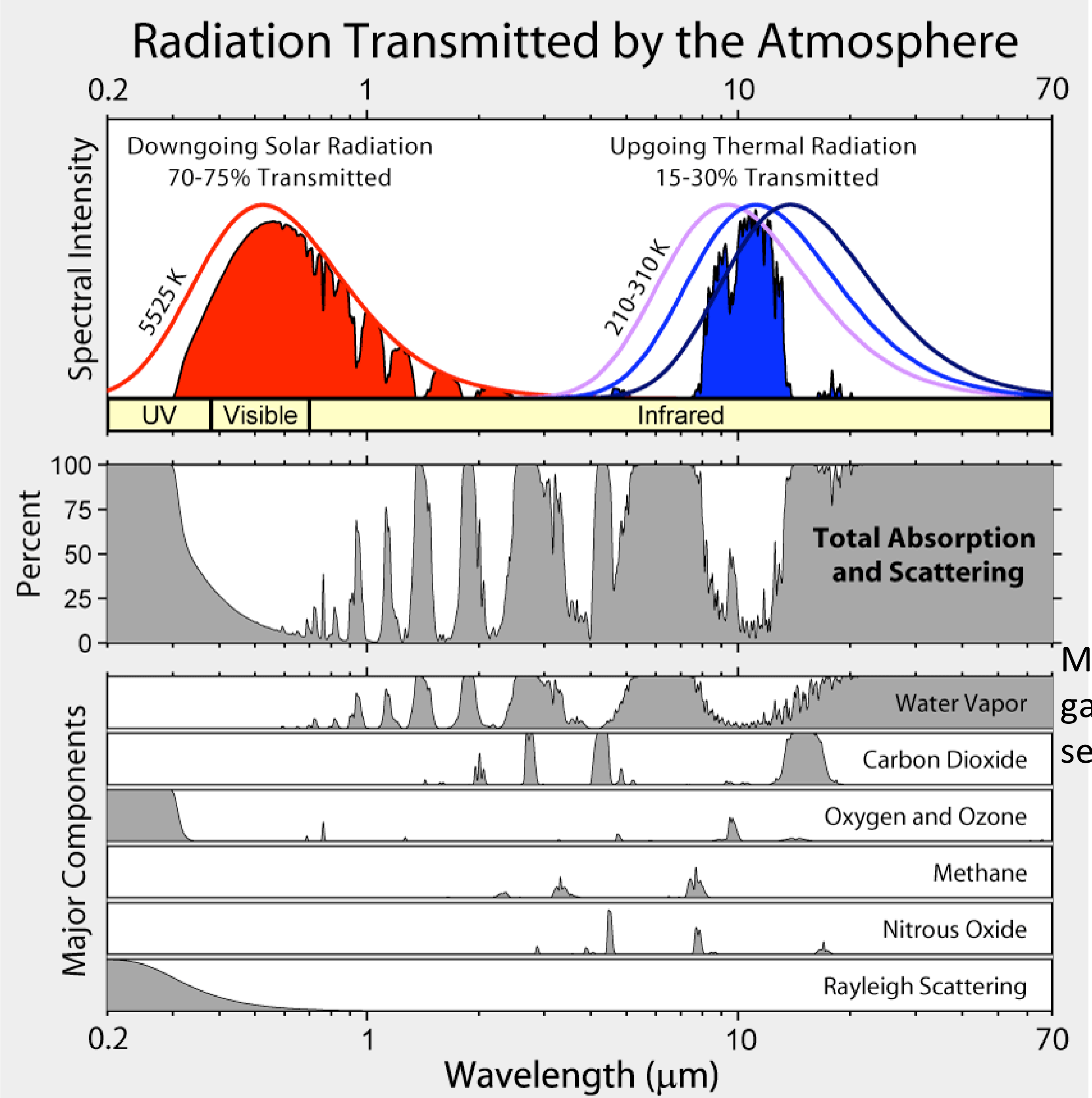
What do we learn from this?



Without greenhouse gases in the atmosphere Earth's surface temperature would be 33℃ colder than they actually are. Earth would be frozen cold like a giant snowball.

Solar Radiation Spectrum





No clouds

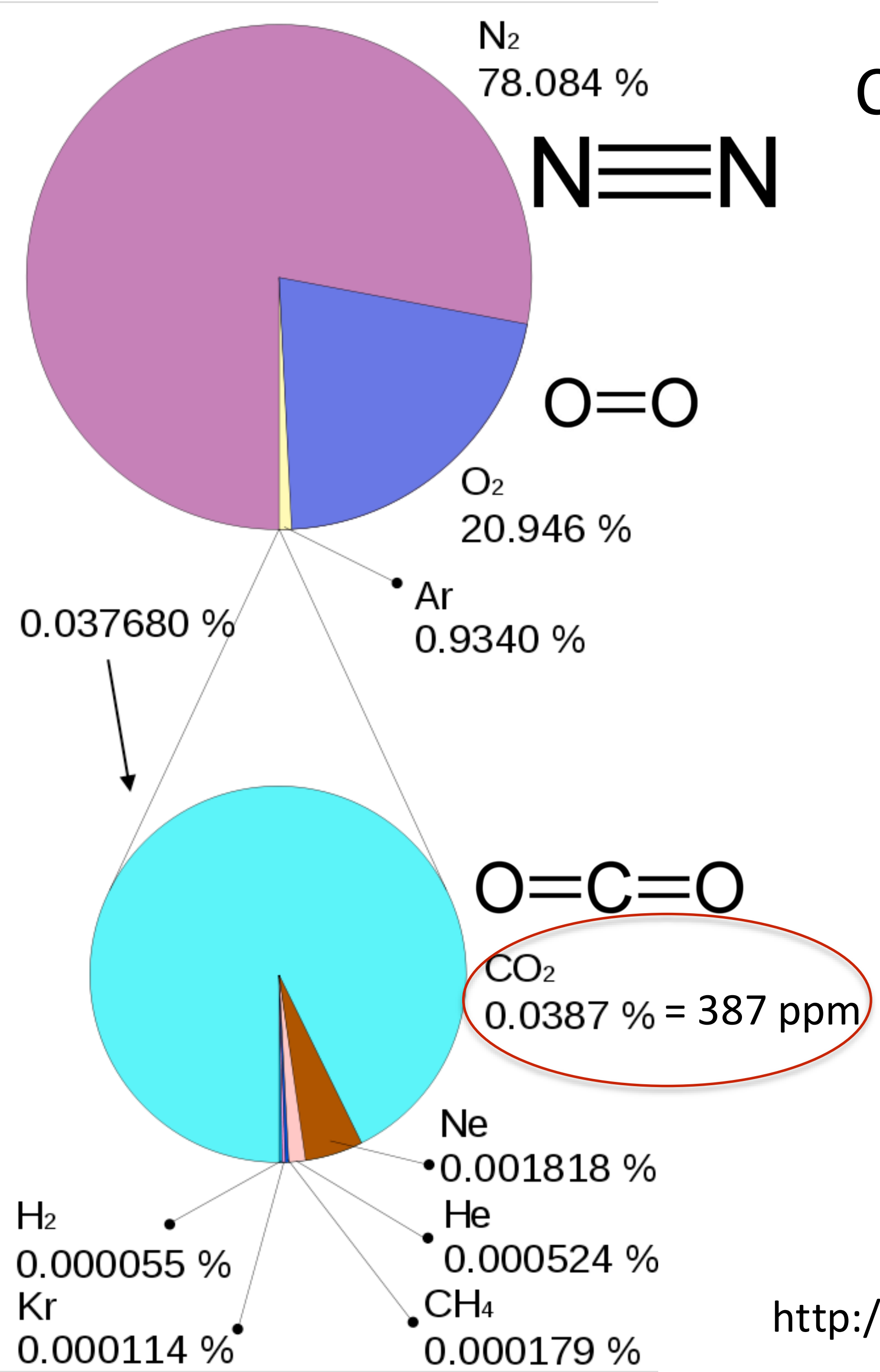
Atmosphere is mostly transparent to shortwave (solar) radiation and mostly opaque to to longwave (terrestrial) radiation.

Most important greenhouse gas is water, second CO_2

http://www.globalwarmingart.com/wiki/File:Atmospheric_Transmi

What is the composition of the atmosphere?

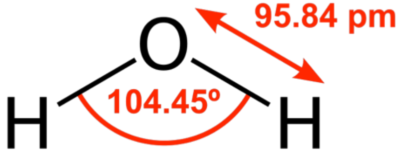
Composition of the Atmosphere



plus

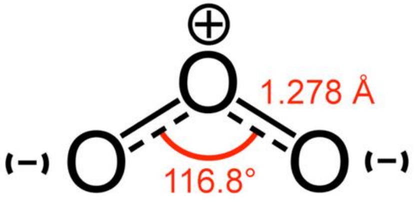
water

H_2O ~ 0.4 %

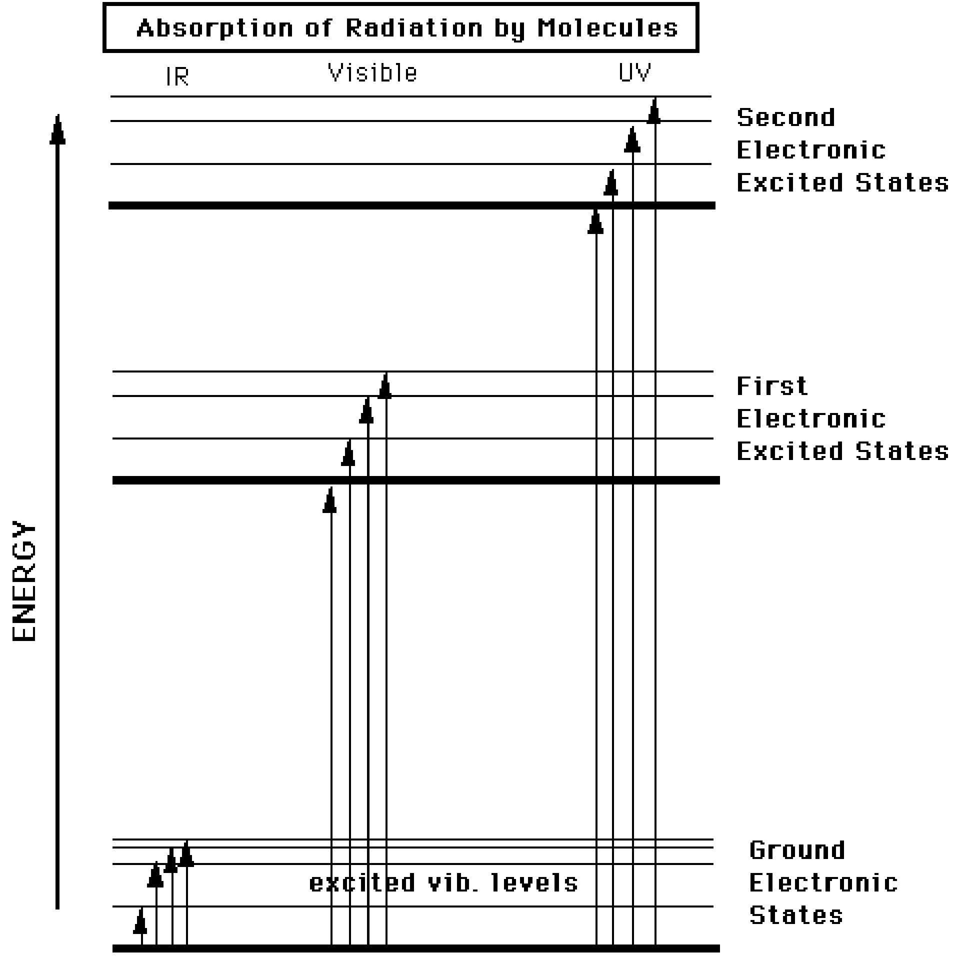


ozone

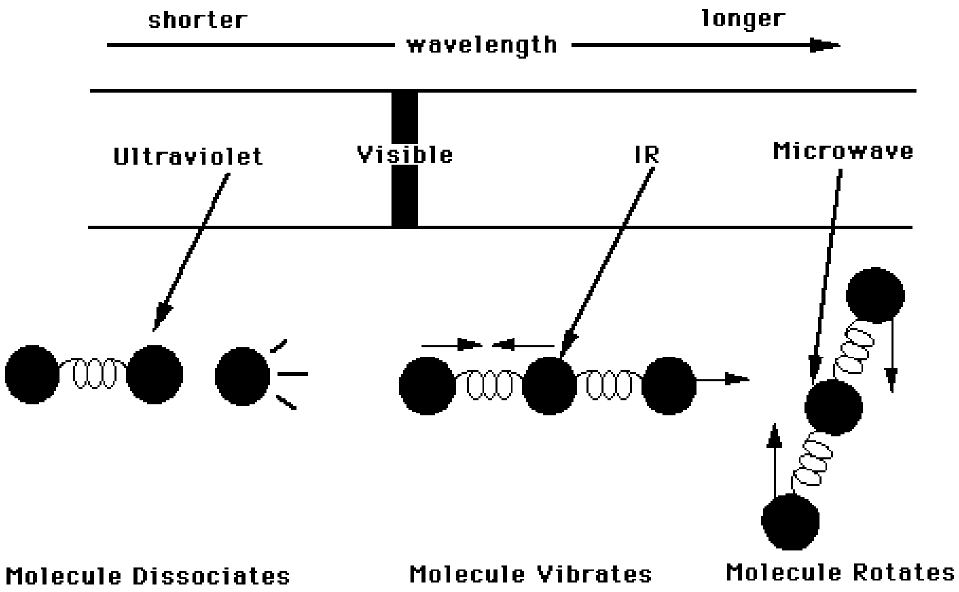
O_3 ~ 0.0-0.07 ppm



http://en.wikipedia.org/wiki/File:Atmosphere_gas_proportions.svg

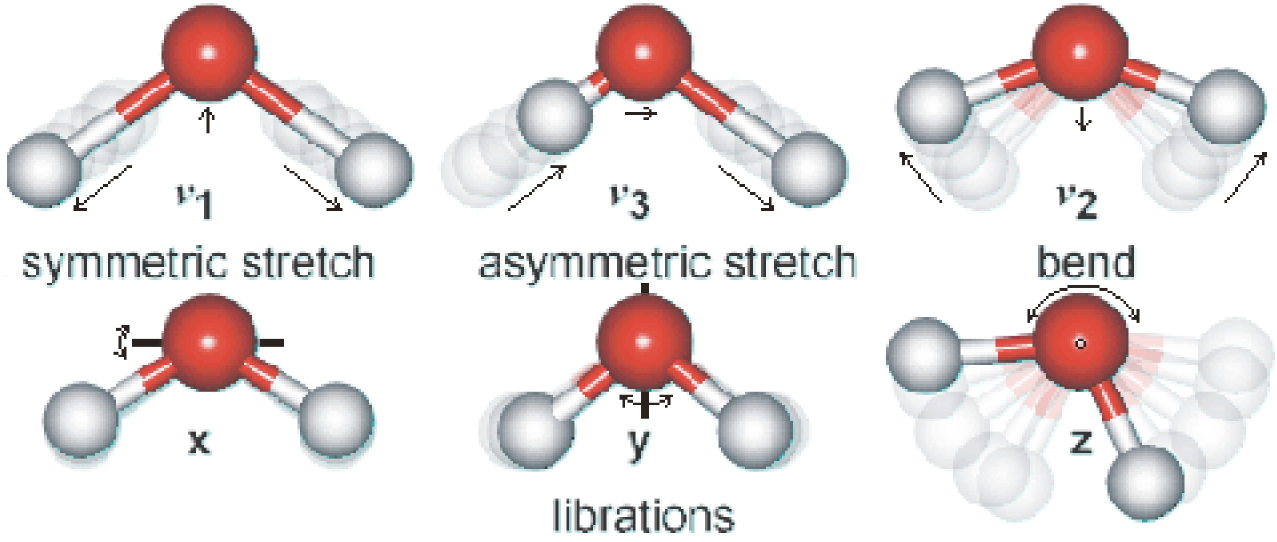


Interaction of electromagnetic radiation with matter



Vibrational Modes of CO_2

Vibrational Modes of H_2O



https://www.youtube.com/watch?v=8dQV-WvR6rA

http://www.wag.caltech.edu/home/jang/genchem/infrared.htm

Why CO2 and not water vapour ?

- Both are greenhouse gases. The difference is control.
- H_2O stays ~10 days condenses, rains out concentration set by temperature → responds to warming
- CO_2 stays centuries never condenses concentration set by emissions → causes warming .
- Warmer air holds more water vapour.

Absorption Coefficients from Spectroscopic Measurements

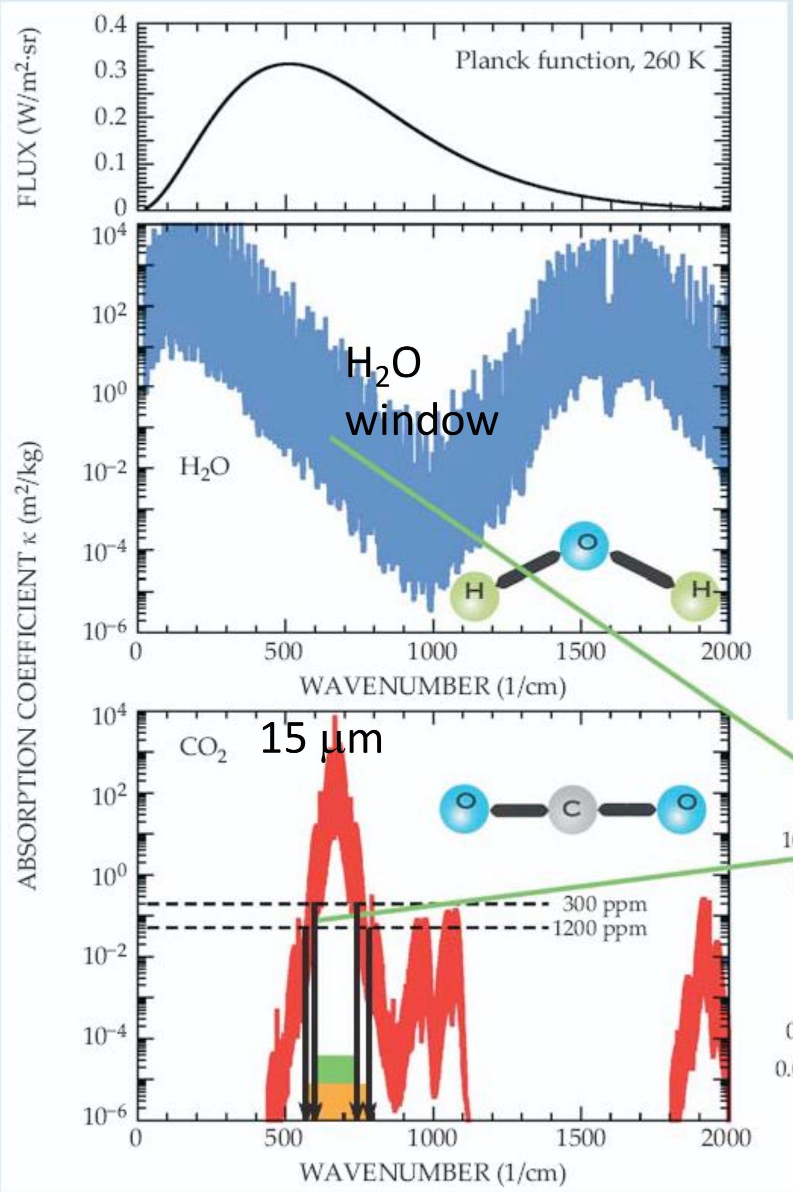
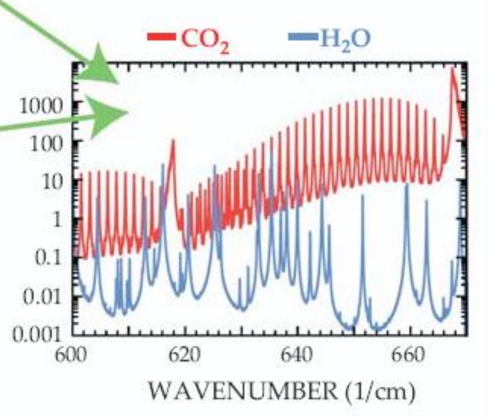
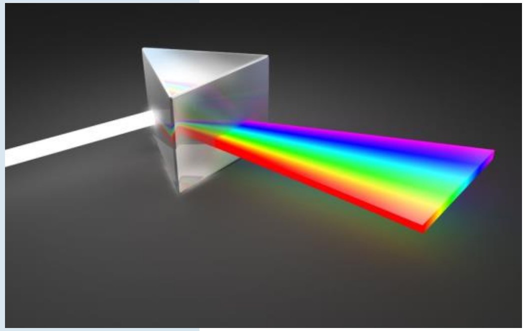


Figure 2. Absorption coefficients for water vapor and carbon dioxide as a function of wavenumber are synthesized here from spectral line data in the HITRAN database. The upper panel gives the Planck function $B(\nu,T)$ for a 260-K surface, which indicates the spectral regions that are important for planetary energy balance. The wavenumber, defined as the reciprocal of the wavelength, is proportional to frequency. If a layer of atmosphere contains M kilograms of absorber for each square meter at the base of the layer, then light is attenuated by a factor $\exp(-\kappa M)$ when crossing the layer, where κ is the absorption coefficient. The horizontal dashed lines on the CO_2 plot give the value of absorption coefficient above which the atmosphere becomes very strongly absorbing for CO_2 concentrations of 300 ppm and 1200 ppm; the green rectangle shows the portion of the spectrum in which the atmosphere is optically thick for the lower concentration, and the orange rectangle indicates how the optically thick region expands as the concentration increases. The inset shows fine structure due to rotational levels.





Plot absorption spectra at:
http://www.spectralcalc.com

Wavenumber = 1/Wavelength

Pierrehumbert 2011 Physics Today, 64 (1), 33pp

Absorption (emission) of radiation by the atmosphere tends to increase (decrease) its temperature.

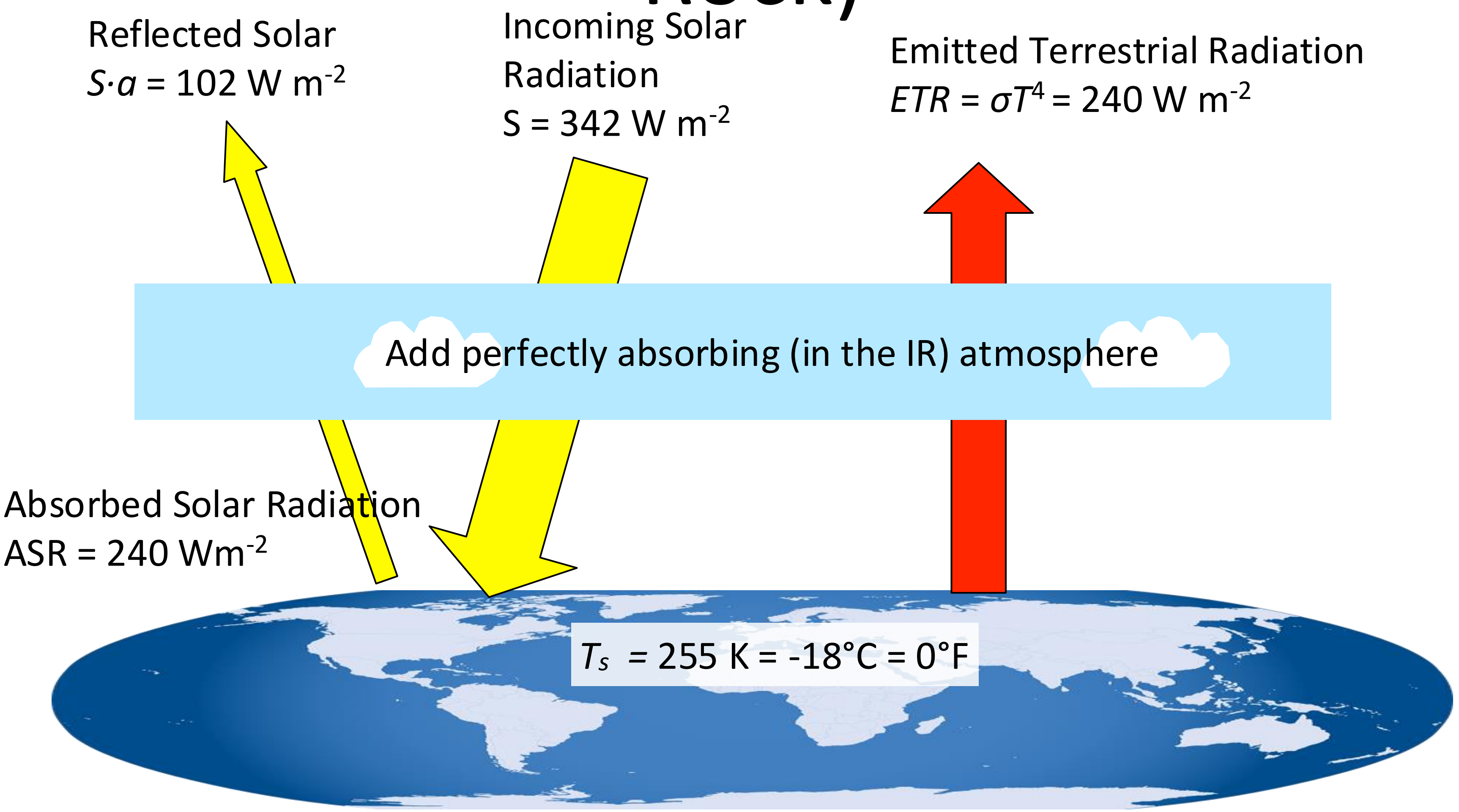
Let’s explore more:

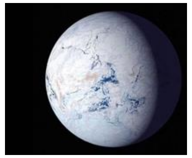
https://phet.colorado.edu/sims/html/greenhouse-effect/latest/greenhouse-effect_all.html

Where we are so far ?

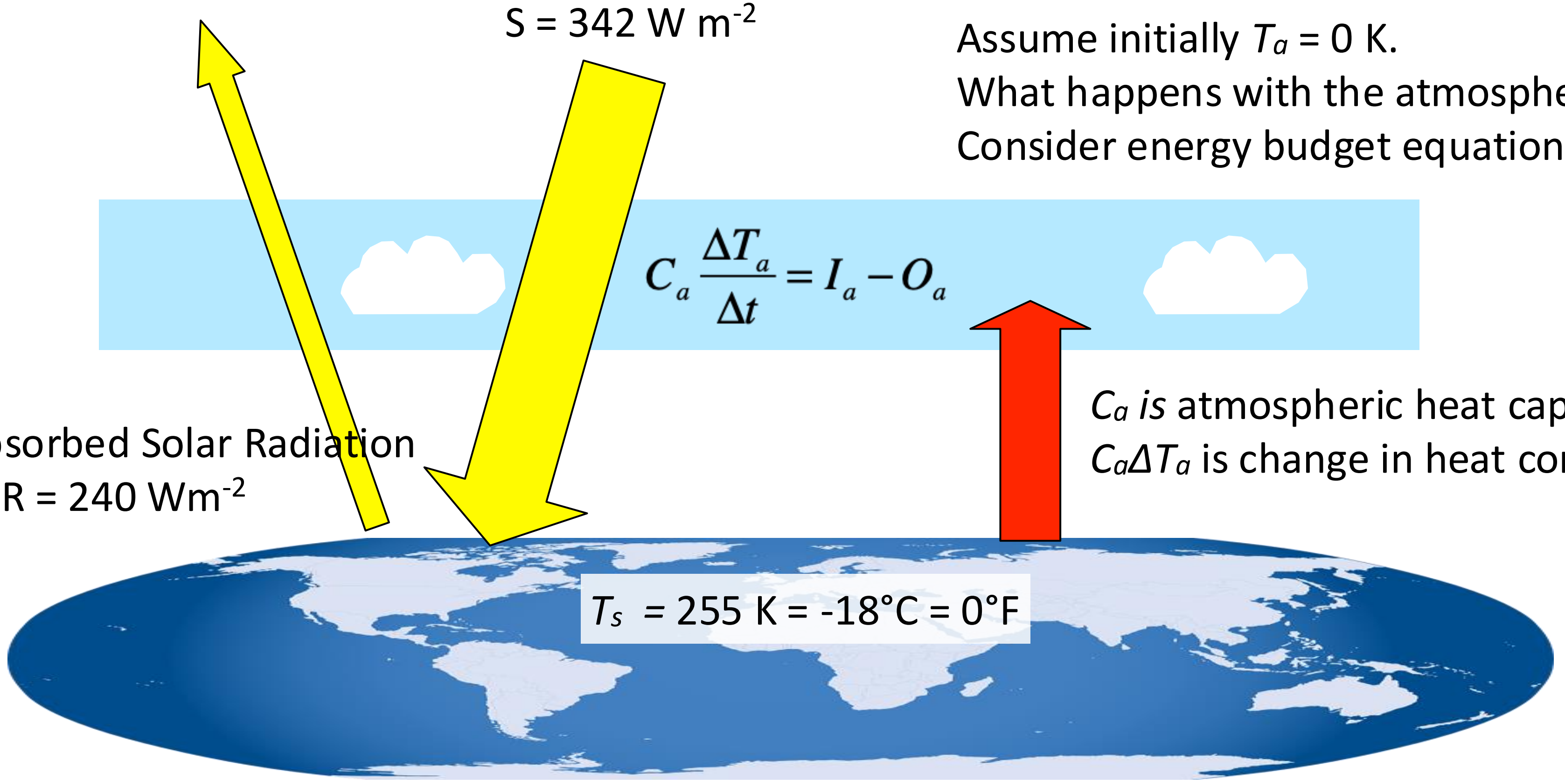
- Budgets: change = in − out. Balance means in = out.
- Everything radiates. σT^4, W/m^2, kelvin only.
- Hot → short wavelengths. Sun 0.5 µm, Earth 10 µm. Two different streas with almost no overlap.
- Bare rock: 240 W/m^2 in → 255 K. Reality: 288 K. Gap = 33°C. 5 99.96% of air is invisible to infrared.
- The 0.04% that isn't: bent or asymmetric molecules.
- Water vapour absorbs more but rains out in 10 days.
- CO_2 stays for centuries.

Energy Balance Model 1 (Bare Rock)



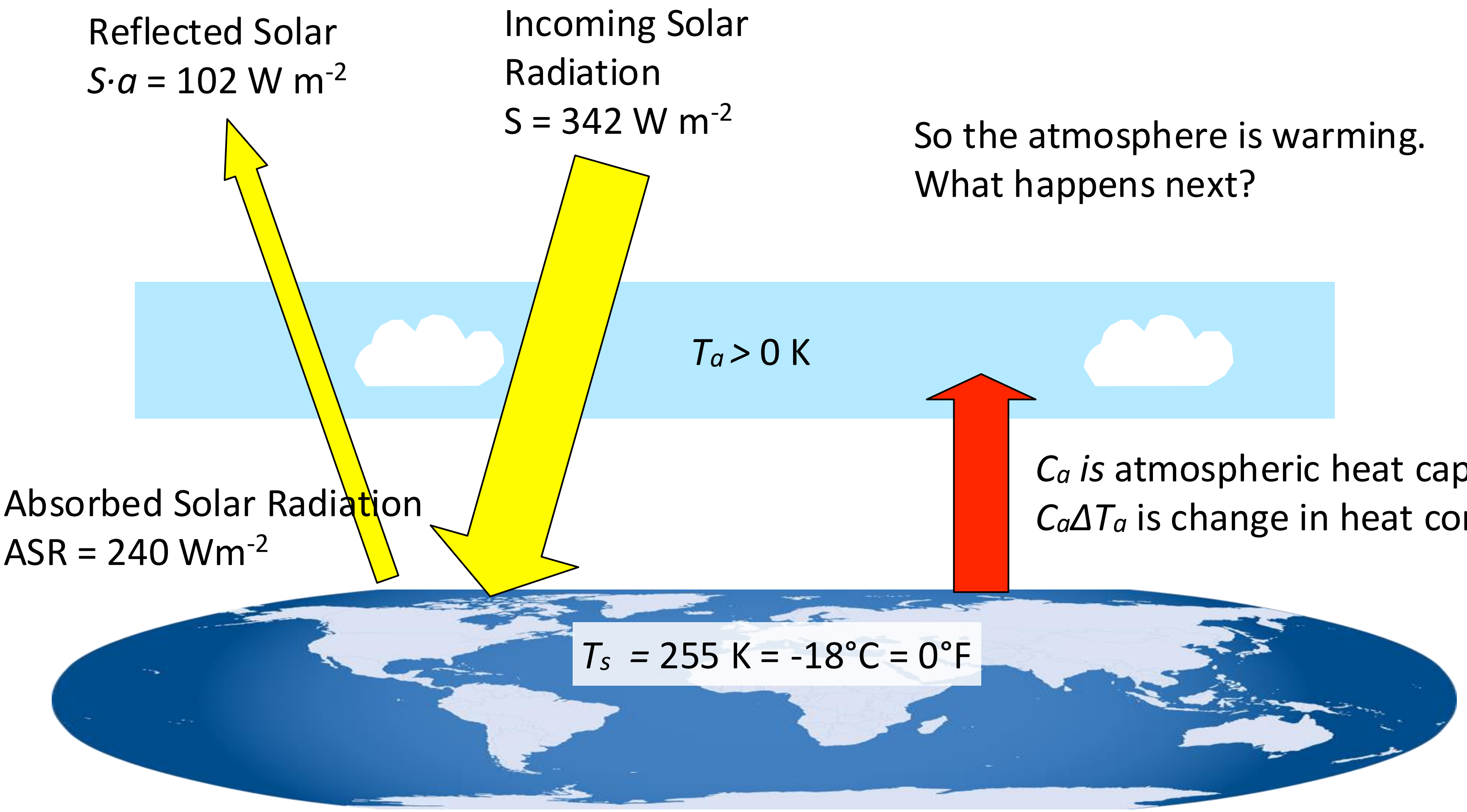


Energy Balance Model 2 (Perfect Greenhouse)



$C_a = m_a c_p = 1 \times 10^4$ J/K.
Calculate the rate-of-change: $\Delta T_a / \Delta t$ = ?

Energy Balance Model 2 (Perfect Greenhouse)

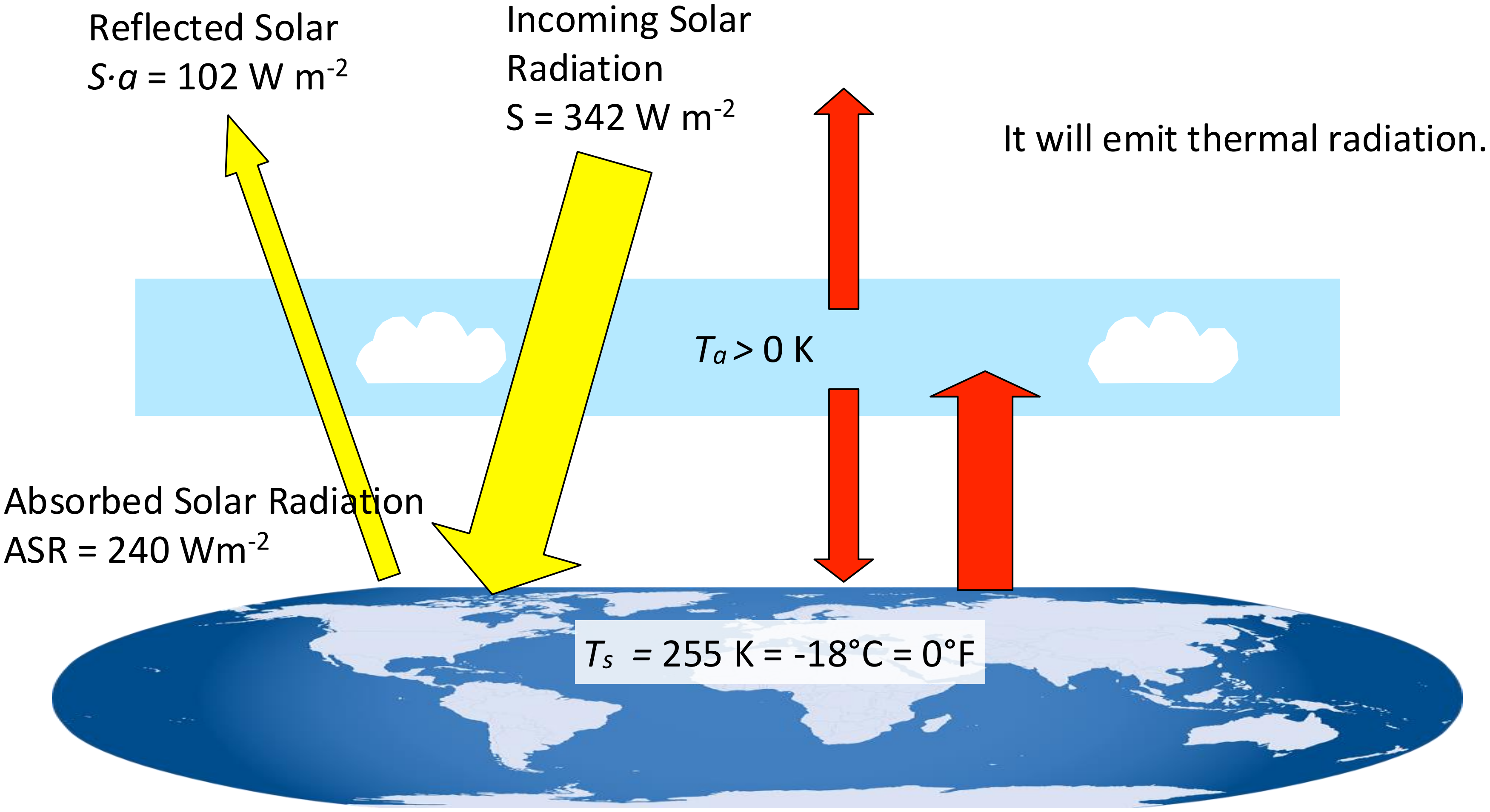


So the atmosphere is warming.
What happens next?

C_a *is* atmospheric heat capacity.
$C_a \Delta T_a$ is change in heat content.

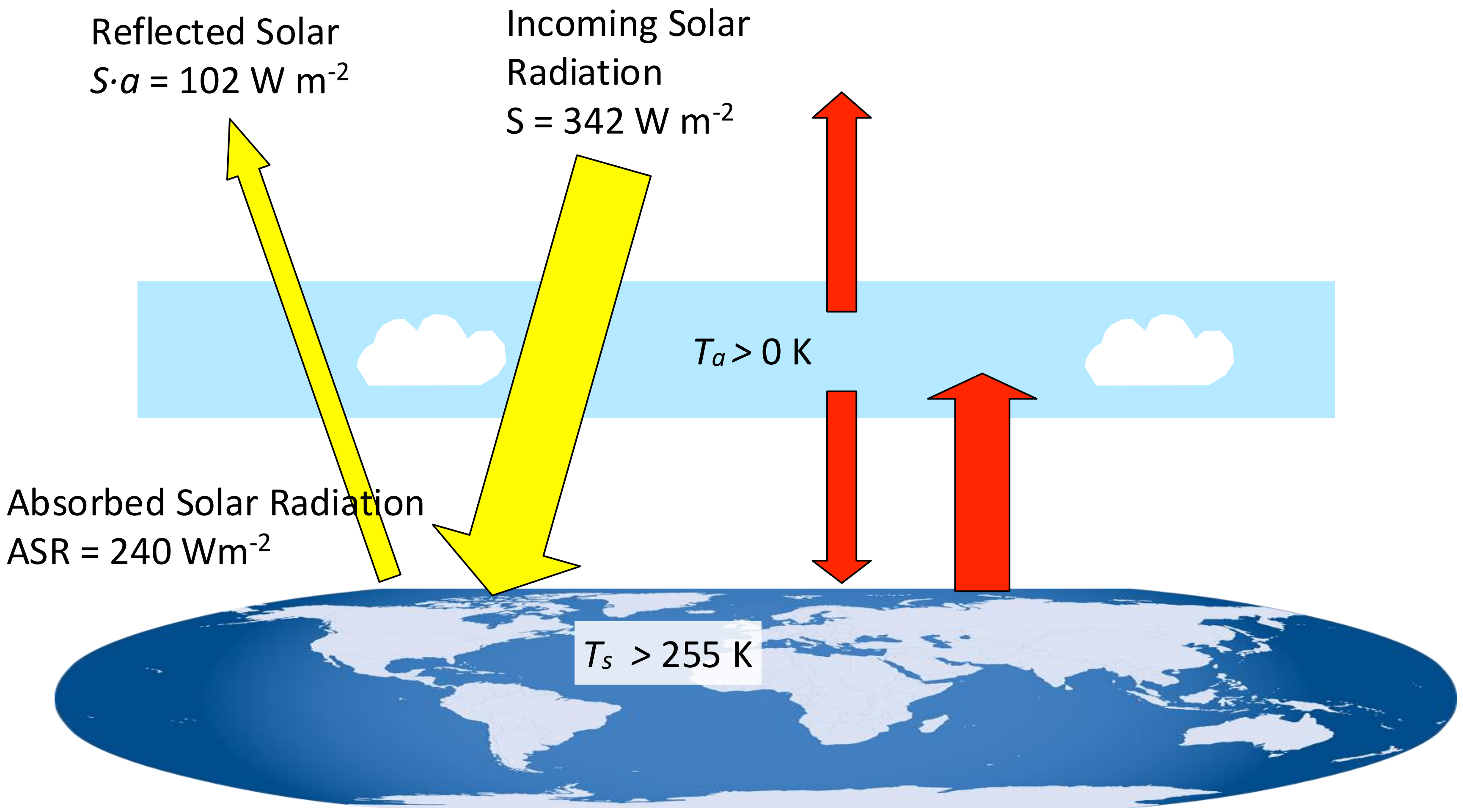
$C_a = m_a c_p = 1{\times}10^4$ J/K.
Calculate the rate-of-change: $\Delta T_a/\Delta t = 0.024\ K/s = 86\ K/h$

Energy Balance Model 2 (Perfect Greenhouse)



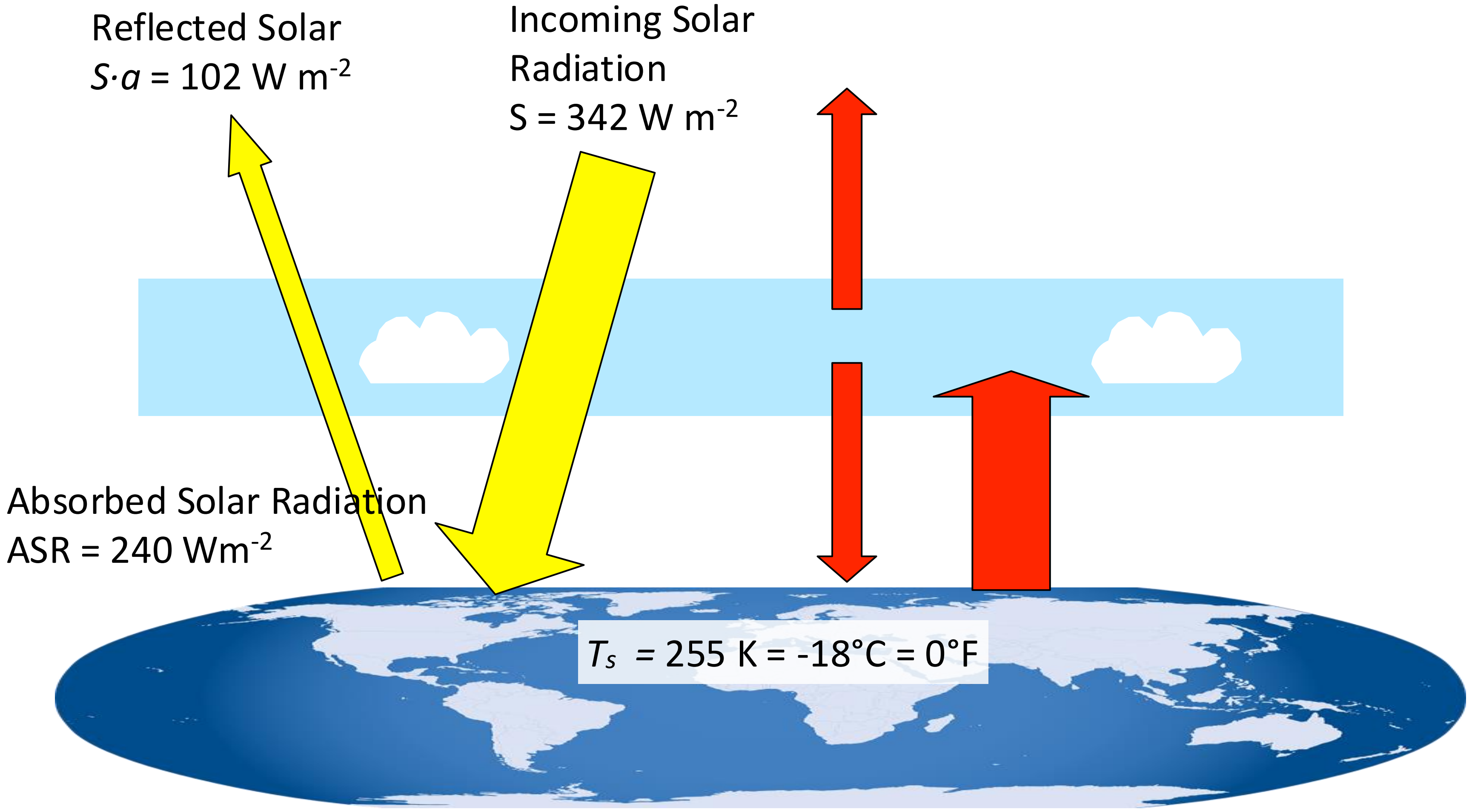
What will happen now with the surface temperature?

Energy Balance Model 2 (Perfect Greenhouse)



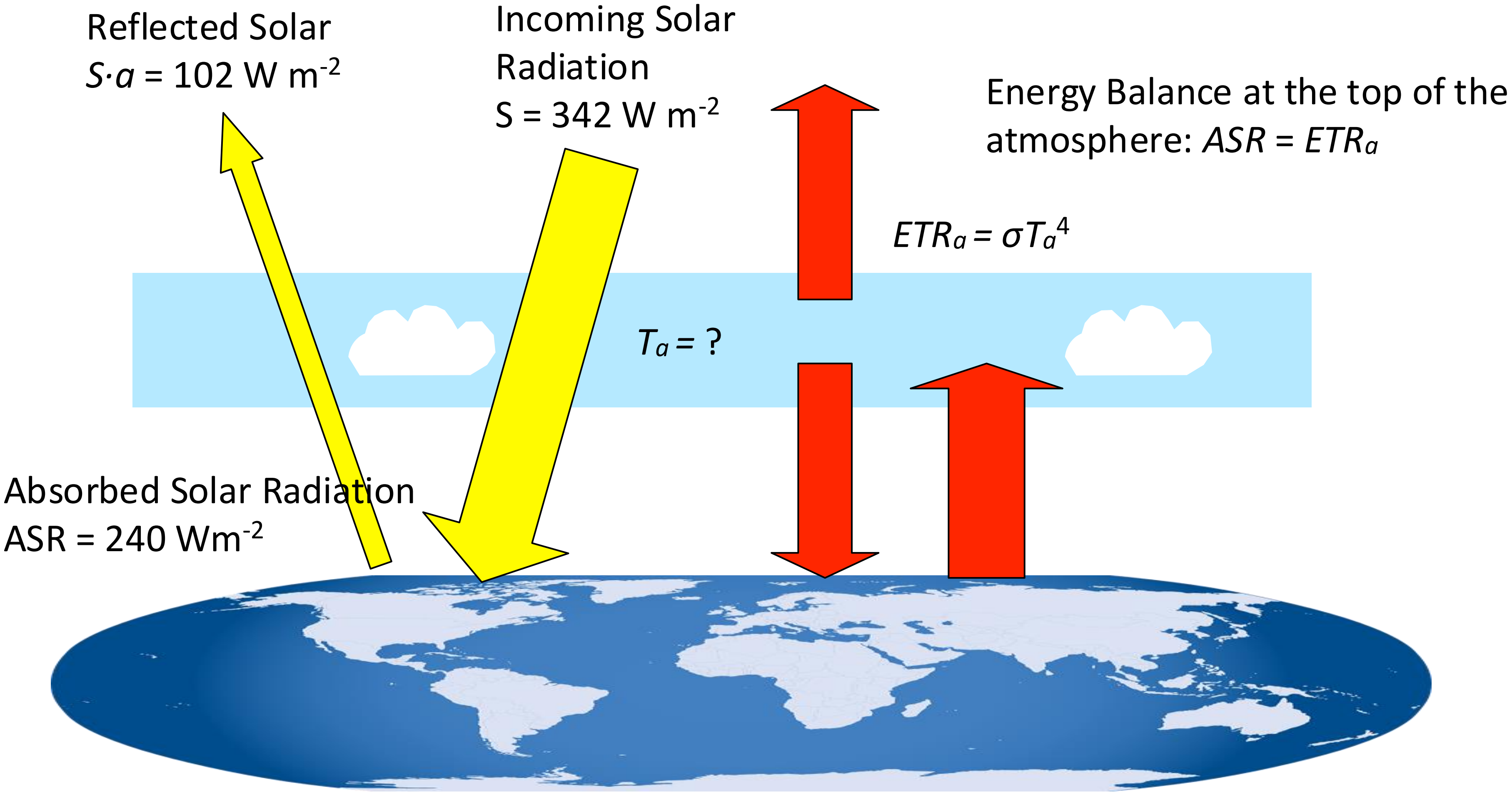
Surface temperature will warm because it receives more energy than it emits.
What happens next with the emitted radiation from the surface?

Energy Balance Model 2 (Perfect Greenhouse)



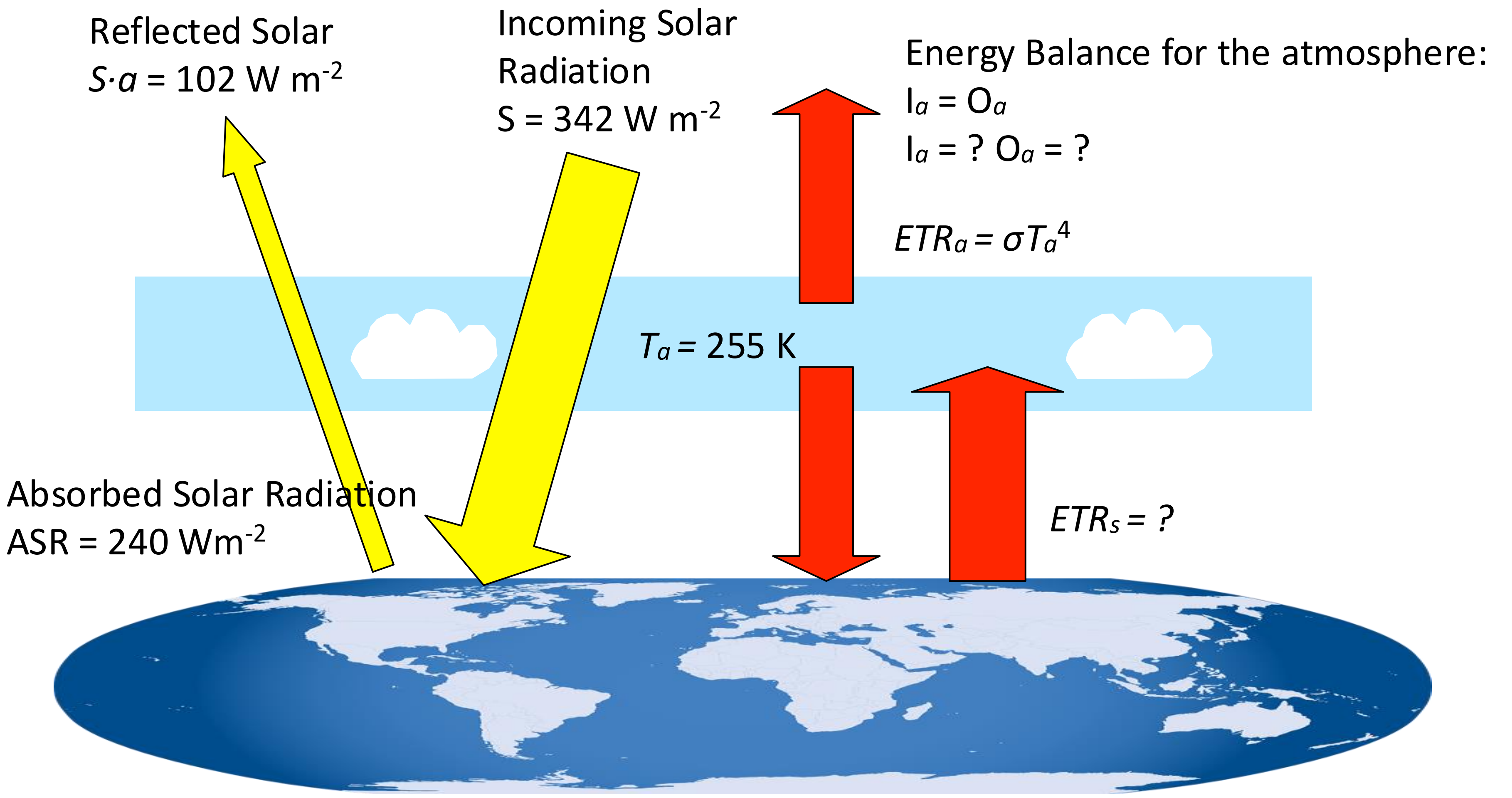
The emitted radiation from the surface will increase, which will increase T_a, …
When will this process stop?

Energy Balance Model 2 (Perfect Greenhouse)



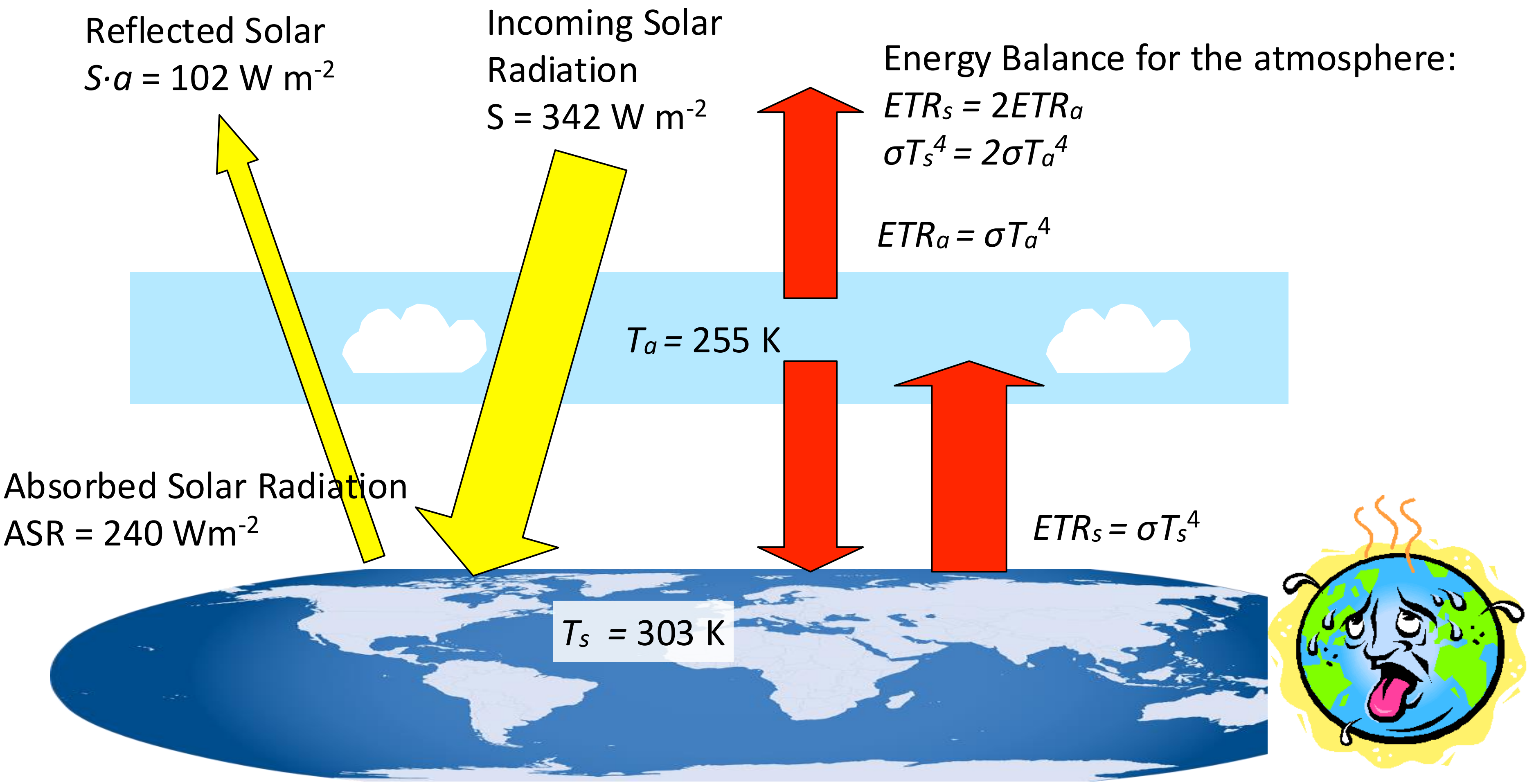
It will stop when the emitted radiation from the atmosphere ETR_a = 240 Wm^{-2}. What will be the atmospheric temperature?

Energy Balance Model 2 (Perfect Greenhouse)



What will be the atmospheric temperature? $T_a = (240\ Wm^{-2}/\sigma)^{1/4}$ = 255 K.
What will be the surface temperature? T_s = ?

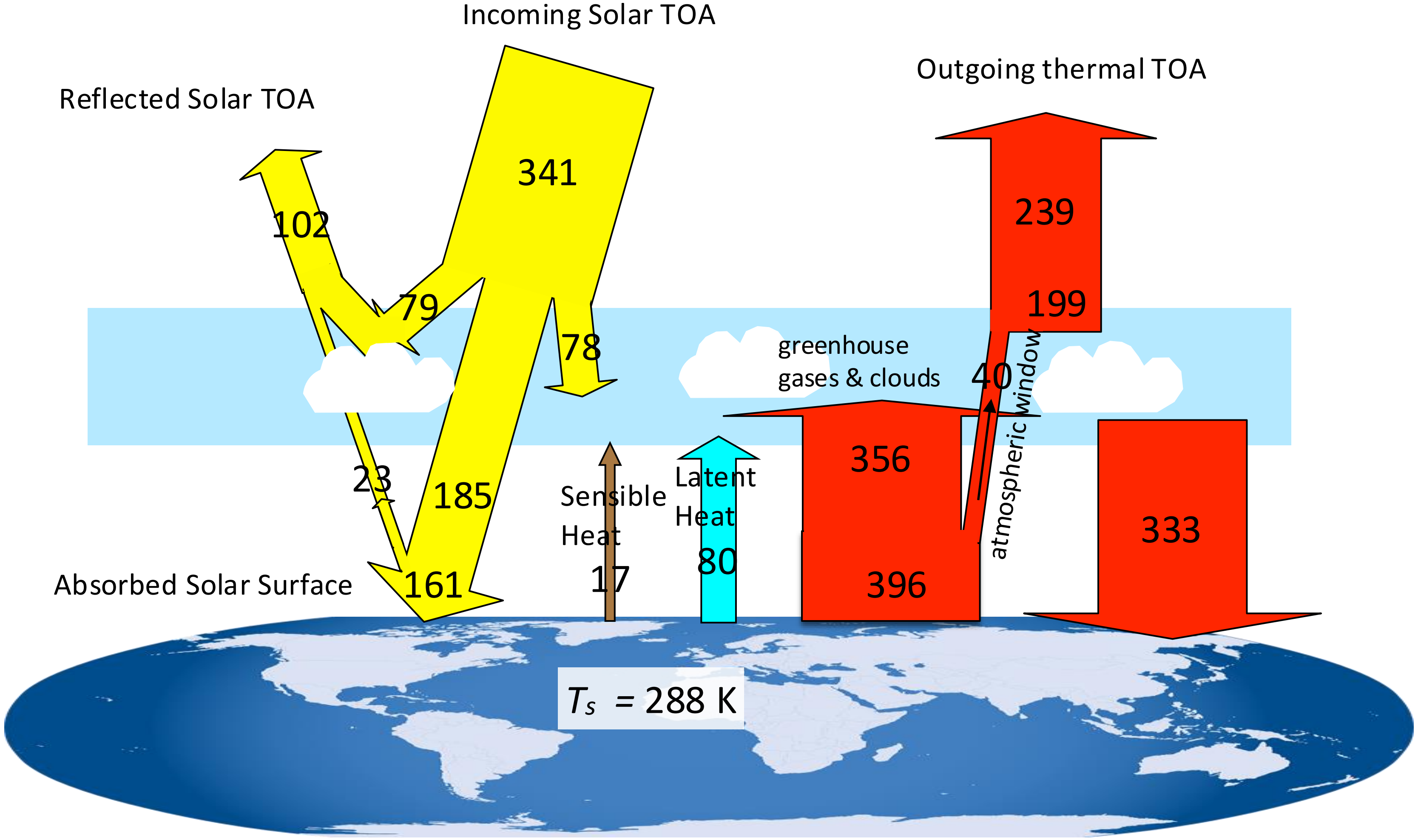
Energy Balance Model 2 (Perfect Greenhouse)



What will be the surface temperature? $T_s = 2^{1/4}T_a = 1.19T_a = 303$ K.

Key Figure

Earth's Energy Budget 3 (Reality)



Warming of the surface by absorbed solar radiation leads to convection, which is rising and sinking air motion. This causes sensible and latent heat fluxes from the surface to the atmosphere, which leads to cooling of the surface and warming of the atmosphere.

Adapted from Trenberth et al. (2009) Bulletin of the American Meteorological Society

Summary

- Greenhouse gases (H_2O, CO_2, CH_4, O_3) in Earth's atmosphere absorb infrared radiation (IR) from the surface, which warms the air
- They re-emit IR radiation in all directions, so that half goes back down, which warms the surface
- This is the greenhouse effect
- The natural greenhouse effect keeps Earth from turning into an ice ball